



Loaded for bear: Bitcoin private wallets, exchange reserves and prices

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ABSTRACT

This study highlights a special feature of cryptocurrency trading and offers information about investor behavior that cannot be observed in traditional financial markets. We find that investors prefer holding bitcoin off exchanges in private wallets and primarily use exchanges to trade. Consequently, bitcoin exchange reserve changes resulting from the movement of bitcoin between private wallets and exchange accounts are negatively related to contemporaneous and future bitcoin returns. Specifically, the transfer of bitcoin on exchanges implies increased selling pressure and the transfer off exchanges implies decreased selling pressure. In turn, flows onto exchanges are triggered by large negative shocks and increased volatility.

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1. Introduction

Private wallets allow investors to store bitcoin off exchanges and therefore separate the storage from the trading - a property of bitcoin that does not exist in traditional financial markets. The separation implies that an investor who holds all bitcoin off exchange and wants to sell bitcoin on an exchange first needs to load bitcoin from a private wallet onto the exchange. The title of this paper “loaded for bear” aims to capture the essence of this mechanism. The term means “prepared for a serious confrontation” and derives from the idea that hunters want the right ammunition for the game they are going after. Hence, investors load their bitcoin (bullets) onto exchange accounts (guns) if they want to sell (shoot), resulting in lower prices. If the selling pressure initiates or deepens a “bear market”, the analogy is even more fitting. Similarly, investors unload their bitcoin (bullets) from their exchange accounts (guns) if they do not want to sell (shoot).¹ The gun analogy also makes a prediction about volatility and risk as investors may con-

sider an environment where guns are loaded as more risky than an environment in which guns are not loaded.

We capture the movement of bitcoin between private wallets and exchange accounts with bitcoin exchange reserves, which increase when investors load bitcoin onto exchanges from private wallets and vice versa.² Our conjecture that exchange reserve changes have implications on price pressure is based on the premise that investors care about storage and move their holdings on and off exchanges. If they did not care, the movement between private wallets and exchange accounts would be random and there would be no negative relation between bitcoin exchange reserves and prices. In other words, our hypothesis is built upon the idea that exchange accounts are a temporary, short-term destination for investors to trade bitcoin on exchanges, while private wallets are used to store bitcoin in the long term. We argue that this is likely the case, mainly due to concerns about bitcoin exchange security since storing bitcoin on exchange accounts implies credit risk, e.g., an exchange becomes insolvent or absconds with customer deposits (Moore and Christin, 2013).³ Moreover, bitcoin ex-

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¹ Another analogy would be a supermarket which puts more umbrellas on the shelves when long spells of rain are expected and takes umbrellas off the shelves when long spells of sunny weather are expected.

² The concept of exchange reserves does not apply to decentralized exchanges which utilize liquidity pools and automated market makers (Aspris et al., 2021; Kozhan and Viswanath-Natraj, 2021; Lehar and Parlour, 2021).

³ A practical example is that clients of Canada's largest cryptocurrency exchange QuadrigaCX were unable to access their US\$250 million funds after the suspicious death of the exchange founder Gerald Cotten who was the only person

changes frequently fall victim to hacking. According to Hackernoon, US\$12.6 billion worth of cryptocurrency has been stolen from exchanges over the last seven years, and the frequency and severity of hacks are increasing despite developments in security procedures.⁴ Thus, investors especially those who intend to hold their bitcoin over a long period may prefer holding bitcoin in private wallets rather than exchange accounts. This indicates that investors have more trust in the blockchain and its private wallets than in centralized exchanges consistent with the fundamental blockchain property “trustless”, i.e. no middleman or central authority needs to be trusted (Nakamoto, 2008).⁵ Our empirical evidence also supports this notion as it shows that an increase of the exchange's bitcoin reserves is associated with an increase in contemporaneous and next-day seller-initiated trades on the exchange.

If investors transfer bitcoin to exchange accounts with the objective to sell them, the changes in bitcoin held on exchanges can be considered a proxy for price pressure which is traditionally measured by order imbalance, i.e., the difference between buyer-initiated and seller-initiated trades. Hence, our study is related to a large body of literature that has documented impacts of order imbalance on returns and volatility (e.g., Berger et al., 2008; Chordia et al., 2002; Dunne et al., 2010; Hasbrouck and Seppi, 2001; Kim and Stoll, 2014; Opschoor et al., 2014; Underwood, 2009). However, we differ from all prior studies which focus on traditional assets such as stocks or foreign currencies in two important ways. First, we study a design - the existence of private wallets - that is unique to bitcoin in particular and cryptocurrencies in general. In traditional financial markets, assets are “stored” in investors' brokerage accounts and investors are unable to transfer them to private wallets. Consequently, investors' intention to sell such assets can only be observed when they actually place an order, which is usually quantified by order imbalance. In the bitcoin design, such intention can be observed even before orders are made through the movement between private wallets and exchange accounts. Hence, this analysis offers information about investor intention which cannot be observed or inferred in traditional financial markets, potentially providing a more timely indicator of price pressure.⁶ While this feature is special and unique for cryptoassets, the transparency of intentions to trade also implies a challenge to market integrity as potential future trades can be front-run.

Since loading onto exchanges reflects investors' intention to sell and not necessarily immediate selling but only selling with a lag, exchange reserve changes may be a better predictor of future prices than order imbalance.

Second, order imbalance data is less readily available in the bitcoin setting. While stocks or bonds are usually listed on a single exchange in a single currency, bitcoin is traded on multiple exchanges and denominated in multiple currencies (including both fiat currencies and cryptocurrencies). As a result, collecting details of bitcoin transactions with other currencies on a large number of exchanges may not be possible due to limited data. For example, according to Coinmarketcap.com there are more than 300 exchanges on which bitcoin is traded - most of them do not make their trade history available to the public. Meanwhile, exchange re-

serves can be easily retrieved from various data vendors such as CryptoQuant, Glassnode or IntoTheBlock.

Our paper is also related to studies attempting to identify factors that affect or predict bitcoin returns and volatility. Most of those studies to date focus on market-based factors such as derivatives (Alexander and Heck, 2020; Alexander et al., 2021; Hoang and Baur, 2020), spot trading volume and liquidity (Balcilar et al., 2017; Bouri et al., 2019; Naeem et al., 2020; Leirvik, 2021), stock, foreign exchange and gold markets (Panagiotidis et al., 2018; 2019), or macro factors such as investor sentiment and attention (Bleher and Dimpfl, 2019; Dastgir et al., 2019; Kristoufek, 2015; Shen et al., 2019), and economic policy uncertainty (Demir et al., 2018; Wang et al., 2020). Apart from those market-based factors which are also widely studied in traditional financial markets, more recent studies analyze blockchain-related characteristics which are unique to cryptocurrencies in general and bitcoin in particular. For example, Aalborg et al. (2019), Hau et al. (2021), and Koutmos (2018) examine the relations between bitcoin returns and the number of transactions and the number of unique addresses on the bitcoin blockchain. Ante and Fiedler (2021) study the bitcoin price reactions to different types of large on-chain transactions, and Liu and Tsyvinski (2020) show that cryptocurrency returns are driven and predicted by a combination of network factors including the number of wallet users, active addresses, transaction and payment counts. Our study particularly contributes to this relatively new strand of literature by investigating price changes as a result of the interaction between bitcoin exchanges, investors and the blockchain. We run a horse race analysis and show that exchange reserves are superior to all other factors studied in prior literature in predicting future bitcoin returns and volatility, both in in-sample and out-of-sample settings.

Utilizing a unique dataset from CryptoQuant that compiles the bitcoin reserves of 20 leading cryptocurrency exchanges, we show that exchanges are involved in a large proportion of bitcoin on-chain transactions. The average is 60% over the sample period from 2016 to 2021, which increased to around 80% in the last three years. However, on average only 12% of total bitcoin in circulation is kept on exchanges and the remaining 88% is stored off exchanges in investors' private wallets.

Consistent with the price pressure hypothesis, we find a negative relation between the daily changes of bitcoin exchange reserves and current and future bitcoin returns. Moreover, supporting our premise on the functional differences between private wallets and exchange accounts, we find that increased bitcoin exchange reserves are associated with contemporaneous and future short-term price falls, while decreased reserves are followed by long-term price rises. This indicates that investors who transfer their bitcoin on an exchange account sell them shortly afterwards, thus creating short-lived selling pressure. On the other hand, transferring bitcoin from exchanges to private wallets lifts the selling pressure and signals a long-term holding strategy, allowing prices to increase in the long term.

Hence, the result is in line with the argument that investors consider exchange accounts as a medium to trade only and use private wallets to hold bitcoin over longer periods. We further show that transfers of bitcoin onto exchanges increase bitcoin volatility over horizons up to one week, which is in line with short-lived selling pressure created by the transfers.

We next turn our attention to what triggers the transfers of bitcoin to exchange accounts. We find that extreme negative bitcoin returns or crashes are associated with increases of bitcoin exchange reserves the next day, implying further selling following market crashes. There is no effect on reserves following extreme positive price movements. Volatility also has an effect on exchange reserves: changes in volatility are positively related to exchange reserve changes. This positive relation is consistent with the “loaded

knowing the passwords. See also <https://www.nytimes.com/2019/12/17/business/gerald-cotten-death-cryptocurrency.html>.

⁴ See <https://hackernoon.com/the-12-6-billion-black-hole-are-your-assets-protected-7ee34d9c810e>.

⁵ See also “The trust machine” (The Economist, October 31, 2015) on <https://www.economist.com/leaders/2015/10/31/the-trust-machine>.

⁶ This is an example of the special features of blockchain-based assets and trading. It is in line with arguments by Yermack (2017) who states that blockchain technology offers new opportunities for financial markets. Interestingly, this special feature may be due to the use of centralized exchanges for decentralized assets. If most trading occurred on decentralized exchanges the differences would be less obvious or non-existent.

for bear" analogy, i.e. loading of the gun in response to increased risk.

Instead of reflecting the transfers of bitcoin from investors' private wallets to exchange accounts, it is also possible that an increase of bitcoin exchange reserves is due to exchanges using their own money to buy (sell) bitcoin. If this was the case, we would observe a positive relation between reserve changes and bitcoin returns due to the buying (selling) pressure from exchanges.⁷ However, since our empirical results reveal a negative relation instead, this effect is small and the variation of bitcoin exchange reserves is mainly due to investors transferring their bitcoin on and off exchanges.

The rest of the paper is structured as follows: Section 2 introduces and describes the data, Section 3 presents the empirical analysis, Section 4 reports the robustness tests, Section 5 presents the comparison with other determinants of Bitcoin returns, Section 6 addresses identification concerns and Section 7 summarizes the main findings and concludes.

2. Bitcoin exchange reserves

Our exchange reserves data are sourced from CryptoQuant.⁸ To calculate daily exchange reserves, the data vendor identifies addresses of various exchanges on the bitcoin blockchain and tracks the amount of bitcoin transferred to and from those addresses. Exchange addresses are identified and updated by frequently depositing, trading and withdrawing bitcoin from targeted exchanges, and tracing the flow of funds on the blockchain from wallets to wallets.⁹ Exchanges included in the calculation are Binance, Bitfinex, Bitfyer, Bithumb, BitMEX, Bitstamp, Bittrex, Bybit, Coinbase Pro, Coinone, FTX, Gate.io, Gemini, Gopax, Huobi Global, Korbit, Kraken, Kucoin, OKEx, and Poloniex. Daily exchange reserves are defined as the aggregate amount of bitcoin held on all exchange addresses by the end of the day, reflecting how much bitcoin investors have deposited to or withdrawn from exchanges during the day. Bitcoin prices at daily and 5-minute frequencies are downloaded from Coinmarketcap.com.¹⁰ The sample period spans 1977 trading days from January 01, 2016 to May 30, 2021.

Figure 1 Panel A depicts the total amount of bitcoin reserves held on all exchanges over time. The reserves display an upward trend from the beginning of the sample until mid 2019 when they peaked at around three billion bitcoins. Subsequently, the exchange reserves gradually decreased to around \$2.4 billion worth of bitcoins by the end of the sample period. It should also be noted that CryptoQuant might not discover all addresses owned by exchanges, so the exchange reserves presented in Fig. 1 can be considered the lower bounds of the actual figures.

Panel B of Fig. 1 exhibits the proportion of exchange reserves to the total amount of bitcoin in circulation.¹¹ The proportion is about 12% on average and exhibits a similar evolution as the level of reserves over time. The relatively small percentage indicates that a large part of bitcoin (88%) is kept off-exchange, in private wallets. Panel E displays the scatter plot of daily log change of bitcoin exchange reserves and bitcoin returns. The two series have a cor-

relation of -0.11 , providing preliminary evidence on the negative impact of exchange reserves on bitcoin returns.

The scatter plot of daily log changes of bitcoin exchange reserves and bitcoin realized volatility in Panel F shows that overall, the two series are positively correlated with a correlation of 0.21. However, when the sample is partitioned into decreased and increased reserve subsamples, exchange reserves and volatility exhibit a V-shaped relation with a steeper slope for positive reserve changes, signalling a possible asymmetric effect.

We next examine what amount of bitcoin transferred on the blockchain is related to exchanges. This is calculated as the daily sum of bitcoin inflows to exchange addresses (i.e. amount of bitcoin received by exchanges) and bitcoin outflows from exchange addresses (i.e., amount of bitcoin sent by exchanges).¹² The data are also provided by CryptoQuant. For a comparison, we further collect the estimated total value of transactions on the bitcoin blockchain from Blockchain.com.¹³

Panel A of Fig. 2 demonstrates that on average exchanges send and receive around 100 thousand bitcoins per day. The number increased significantly in 2017 and became relatively stable recently. In contrast, the total amount of bitcoin transferred on the blockchain markedly decreased after 2018. A possible reason is the increasing popularity of exchanges which allows investors to trade bitcoin off-chain with much lower transaction fees, leading to the observed reduction in direct on-chain trading between two parties. Panel B shows that the ratio of the exchange-related transaction amount to the total transaction amount has increased substantially since 2018, indicating that exchanges are primarily used to buy and sell bitcoin. Over-the-counter and private transactions, which represent the rest, usually only account for less than 20% of total on-chain transaction value, which is consistent with the conclusion by Baur et al. (2018) that bitcoin is rather a speculative asset than a medium of exchange.

Panel C shows that on average nearly 100 thousand bitcoins are transferred between non-exchange addresses per day, which is equivalent to about 500 million US dollars as shown in Panel D. Assuming that exchanges are not involved in illegal trading, the numbers presented in Panels C and D can be inferred as the upper bounds of daily illegal trading conducted by bitcoin users. Although such upper bounds denominated in bitcoin significantly decreased during the last three years, the numbers in US dollars exhibit an opposite trend due to the increase of bitcoin prices. While our average upper bound estimate is US\$4 billion per month and thus lower than the US\$10 billion estimate of Foley et al. (2019) for the same period (January 2016 to April 2017),¹⁴ our estimate based on the full and longer period is about US\$15 billion (US\$500m per day $\times$ 30 days) exceeding the estimates reported in Foley et al. (2019).

In general, the descriptive analysis shows that a significant fraction of bitcoin investors store their wealth off exchanges and in their private wallets. Moreover, exchanges handle the majority of bitcoin transactions implying that investors predominantly use exchanges to buy or sell bitcoin.

3. Econometric analysis

This section presents the econometric analysis of the effect of exchange reserve changes on bitcoin returns and volatility, and triggers of such reserve changes.

⁷ If an exchange uses its main wallet to buy and sell Bitcoin, any buying would lead to increased reserves and any selling would lead to decreased reserves resulting in a positive relationship.

⁸ <https://www.cryptoquant.com/>.

⁹ Please refer to the technical document provided by CryptoQuant at <https://bit.ly/3nfcdBt> for more details about the process.

¹⁰ Coinmarketcap.com is a leading source of cryptocurrency price and volume data which is widely used in the cryptocurrency literature (e.g., Griffin and Shams, 2020; Hårdle et al., 2020; Liu and Tsyvinski, 2020).

¹¹ Data on the amount of circulating bitcoin are collected from Blockchain.com at <https://www.blockchain.com/charts/total-bitcoins>.

¹² Daily exchange reserves used in the main analysis is calculated as the sum of yesterday's reserve and today's inflow minus today's outflow.

¹³ <https://www.blockchain.com/charts/estimated-transaction-volume>

¹⁴ See Fig. 6 (Foley et al., 2019).

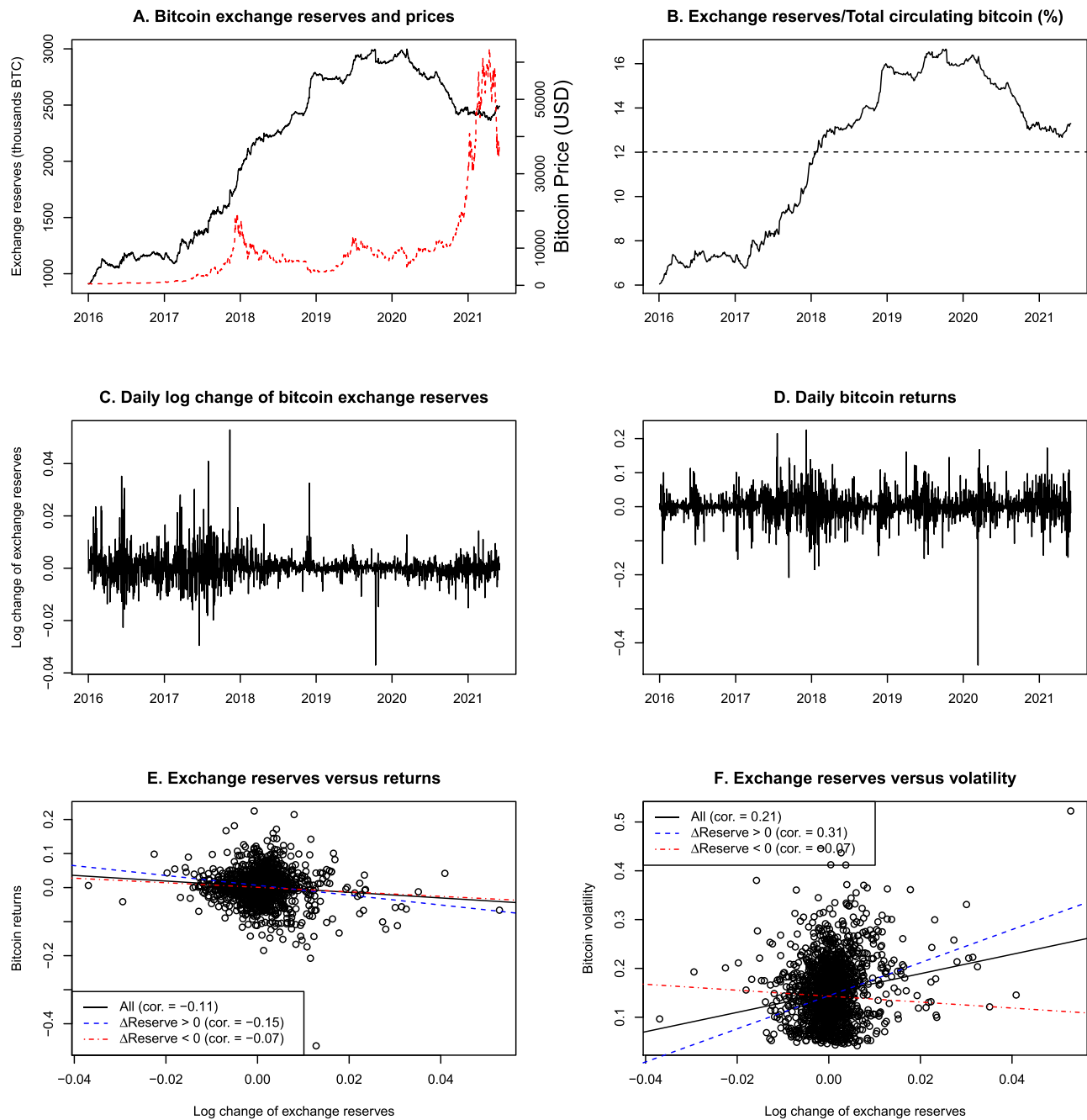


Fig. 1. Bitcoin exchange reserves, returns and volatility.

Panel A displays the total amount of daily bitcoin exchange reserves (the black line) and bitcoin prices (the red dashed line). Panel B displays the ratio (in percentage) of bitcoin exchange reserves to total circulating bitcoin over time, with the daily average represented by the dashed line. Panel C and D respectively display the daily log change of bitcoin exchange reserves and daily bitcoin returns. Panels E and F respectively present the scatter plots of daily bitcoin returns and realized volatility against the log change of daily bitcoin exchange reserves. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.1. Effects of bitcoin exchange reserves on bitcoin returns

3.1.1. Baseline regression

We conjecture that when investors load bitcoin onto an exchange, they are likely to sell them in the very near future lowering the price in the short-term. On the other hand, investors who transfer their bitcoin from exchanges to their private wallets expect the price to increase in the long term.

We use the following regression model (1) to formally examine the relation between bitcoin exchange reserves and bitcoin prices:

$$ret_{[t+k,t+h]} = \alpha + \beta \Delta Reserve_t + \sum \gamma_i Control_{i,t} + \varepsilon_{[t+k,t+h]} \quad (1)$$

where $\Delta Reserve_t$ is the log change of bitcoin reserves on all exchanges on day t , and $ret_{[t+k,t+h]}$ denotes bitcoin returns from day $t+k$ to day $t+h$. Bitcoin returns are calculated as the log change of prices over the horizon.¹⁵ We adopt various values of (k, h) to examine the daily contemporaneous relation ($k = h = 0$), the daily

¹⁵ Since both exchange reserves and prices are non-stationary, the log-change transformation is applied to obtain stationary time series. We use the log-change transformation for both series for consistency, however, the results are qualitatively unchanged if we use exchange netflows calculated as the absolute daily change of bitcoin exchange reserves (or equivalently the difference between daily inflows and outflows).

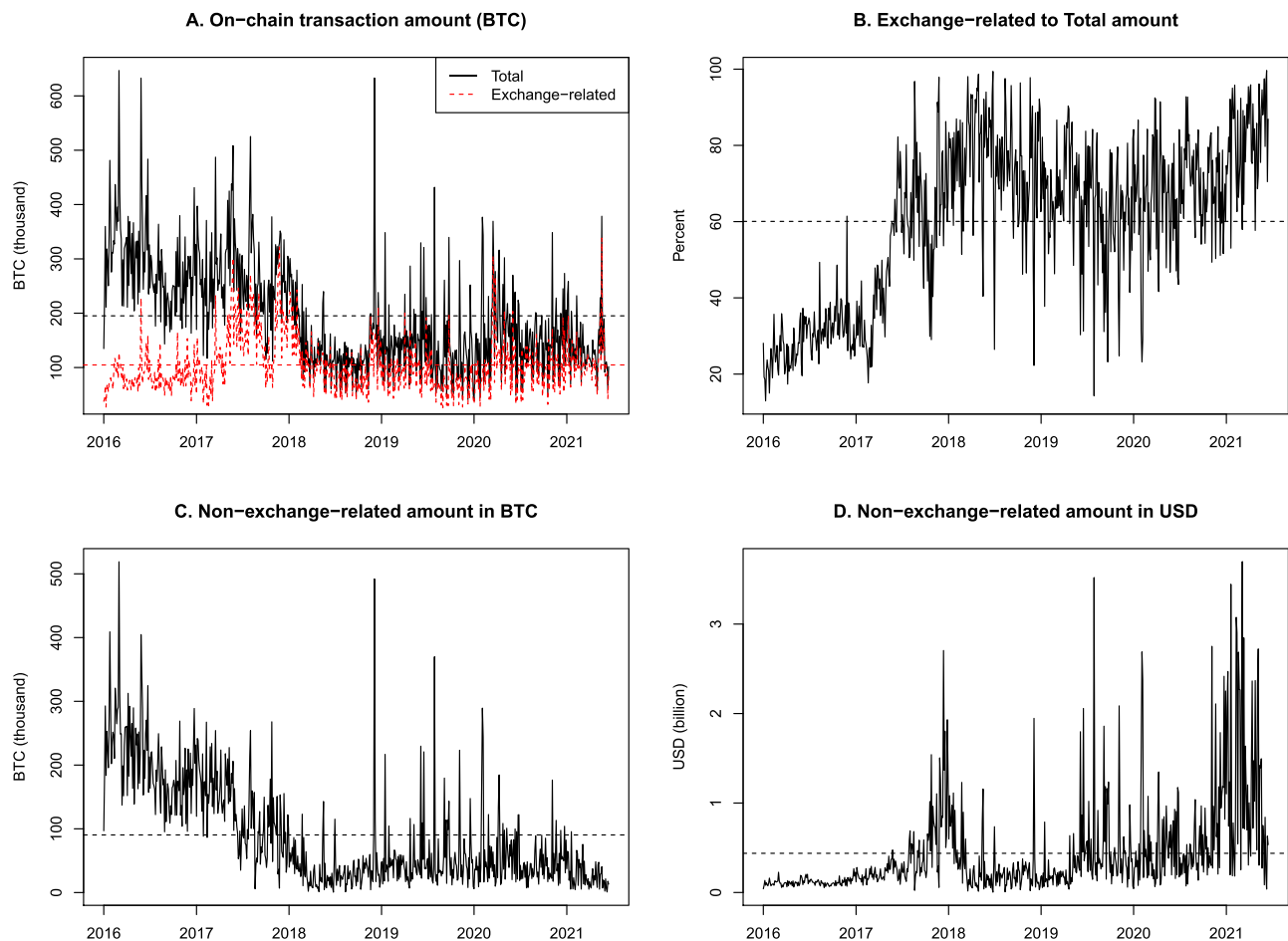


Fig. 2. Exchange-related transactions on bitcoin blockchain.

Panel A displays the exchange-related amount of bitcoin transferred on the blockchain where exchanges are either senders or receivers of bitcoin (the red dashed line) and the total amount of bitcoin transferred on the blockchain by all users (the black line). Panel B displays the ratio (in percentage) of the exchange-related amount over the total amount. Panel C and D respectively displays the amount and value in US dollars of bitcoin transactions on blockchain that are unrelated to exchanges. In all panels, the horizontal dashed lines represents the daily averages. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

lead-lag effect ($k = h = 1$), and the effect on long-term returns including one-week ($k = 1, h = 7$) and one-month ($k = 1, h = 30$). $Control_{i,t}$ denotes a matrix of control variables, including past returns (ret_t and $ret_{[t-21,t-1]}$), the natural log of daily bitcoin trading volume on exchanges in US dollars ($logVolume_t$), and the natural log of daily market capitalization in US dollars ($logMktCap_t$). Data on bitcoin trading volume and market capitalization are collected from Coinmarketcap.com.

Table 1 presents the descriptive statistics of the variables used in the model.

The OLS regression results of model (1) are presented in Table 2. We calculate t -statistics using the Newey-West standard errors with six lags.¹⁶ Consistent with our hypothesis, the coefficients of $\Delta Reserve$ are negative and statistically significant in columns (1) and (2), indicating that bitcoin exchange reserves negatively affect bitcoin returns, both contemporaneously and subsequently. The result is different from Ante and Fiedler (2021), possibly because the study only considers large transfers (500 bitcoins or more) onto hot wallets of two exchanges Bitfinex and Binance, and do not consider the effects of transfers off exchanges.¹⁷

¹⁶ We follow Greene (2003) to set the number of lags as the integer part of $T^{1/4}$ where T is the number of observations. However, the results remain qualitatively similar when we vary the number of lags from 3 to 12.

¹⁷ Ante and Fiedler (2021) find that large deposits into Bitfinex's and Binance's hot wallets are followed by positive returns in the next 15 days.

Table 1

Descriptive statistics.

This table reports descriptive statistics of variables used in this study. $\Delta Reserve$ is the log change of daily bitcoin exchange reserves. $ret_{[t+k,t+h]}$ is bitcoin returns from day $t+k$ to day $t+h$. $RV_{[t+k,t+h]}$ is realized volatility from day $t+k$ to day $t+h$, which is calculated as the squared root of the sum of squared 5-minute returns during the period.

	Mean (1)	Median (2)	Std.Dev (3)	Min (4)	Max (5)
$\Delta Reserve$	0.0005	0.0003	0.0052	-0.0369	0.0528
ret_t	0.0022	0.0025	0.0410	-0.4932	0.2377
$ret_{[t+1,t+7]}$	0.0045	0.0052	0.0562	-0.4866	0.3544
$ret_{[t+1,t+30]}$	0.0721	0.0598	0.2399	-0.9070	1.0788
$ret_{[t-21,t-1]}$	0.0466	0.0397	0.1933	-0.6671	0.8424
RV_t	0.0260	0.0200	0.0216	0.0017	0.2731
$RV_{[t+1,t+7]}$	0.0736	0.0607	0.0507	0.0067	0.3495
$RV_{[t+1,t+30]}$	0.1585	0.1322	0.0897	0.0222	0.4517
$logVolume$	22.0148	22.5736	2.3261	17.1659	25.5624
$logMktcap$	25.0906	25.4768	1.3602	22.4274	27.8019

However, bitcoin exchange reserves in general do not predict long-term returns, as shown by statistically insignificant coefficients of $\Delta Reserve$ in columns (3) and (4).

It is possible that the negative relation between reserve changes and future returns merely reflect the momentum effect (Liu and Tsyvinski, 2020; Liu et al., 2022), that is, negative returns are followed by future negative returns and transfers onto exchanges. To rule out this possibility, we control for the momentum effect by

Table 2

Effects of bitcoin exchange reserves on bitcoin returns.

This table reports regression estimates of bitcoin concurrent and future returns on log change of daily bitcoin reserves ($\Delta Reserve$) and control variables. $ret_{t+k,t+h}$ is bitcoin returns from day $t+k$ to day $t+h$. $logVolume$ is the the natural log of the daily bitcoin trading volume on exchanges in US dollars. $logMktCap$ is the natural log of daily bitcoin market capitalization in US dollars. t -statistics are adjusted using Newey-West standard errors and reported in parentheses. *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

	Dependent variable:			
	ret_t (1)	ret_{t+1} (2)	$ret_{t+1,t+7}$ (3)	$ret_{t+1,t+30}$ (4)
$\Delta Reserve_t$	-0.844*** (-3.273)	-0.374** (-2.228)	-0.949 (-1.283)	-1.381 (-0.569)
ret_t		-0.041 (-1.378)	0.045 (0.706)	0.107 (0.748)
$ret_{t-21,t-1}$	0.011** (2.141)	0.012** (2.209)	0.050 (1.061)	0.107 (0.836)
$logVolume_t$	-0.002 (-1.558)	0.001 (1.154)	0.007 (0.723)	0.025 (0.690)
$logMktCap_t$	0.003 (1.413)	-0.003 (-1.367)	-0.016 (-0.880)	-0.061 (-0.776)
Constant	-0.036 (-1.138)	0.049 (1.551)	0.273 (1.015)	1.050 (0.863)
Observations	1977	1976	1970	1947
R^2	0.016	0.007	0.014	0.022
Adjusted R^2	0.014	0.005	0.011	0.020

including three-week past returns ($ret_{t-21,t-1}$) as a control variable in all predictive models. The results show that the coefficient of $ret_{t-21,t-1}$ is statistically significant in columns (1) and (2), confirming the momentum effect. However, the coefficients of $\Delta Reserve$ remain significant even after controlling for $ret_{t-21,t-1}$. Furthermore, analysis in Section 3.3 shows that negative returns in general do not trigger transfers of bitcoin onto exchanges; only extreme negative returns do. Thus, it can be concluded that the observed relation is independent of the momentum effect.

It is widely documented that due to overlapping observations in the dependent variable in long-horizon predictive models, the estimated coefficients are biased. While Newey-West standard errors may partly alleviate this issue, they are still downward biased (Boudoukh et al., 2008; Ang and Bekaert, 2007). Thus, we further adjust the standard errors using two alternative approaches proposed by Hodrick (1992) and Boudoukh et al. (2019). In addition, Stambaugh (1999) notes that if the independent variable exhibits strong autocorrelation, the estimated coefficient of the one-period-ahead predictive model could be biased. Since the autocorrelation of $\Delta Reserve$ in our sample is relatively small (e.g., 0.083 for the first-order), we expect the issue to be less severe than what has been documented in prior literature which usually studies highly persistent regressors such as dividend yields (Boudoukh et al., 2008). To be sure, we further implement the procedure described in Stambaugh (1999) to correct for the potential bias. Details of the three approaches are presented in Appendix A.

As shown in Table 3, the results are qualitatively unchanged after correcting for the bias. Specifically, the predictive power of $\Delta Reserve$ on one-day-ahead bitcoin returns ret_{t+1} remains statistically significant after correcting for the bias, and the coefficients of $ret_{t+1,t+7}$ and $ret_{t+1,t+30}$ remain statistically insignificant for the majority of corrections.

Due to its robustness, we calculate t -statistics based on Hodrick (1992) in all subsequent analyses where applicable (i.e., in long-horizon predictive models). Otherwise, Newey-West standard errors with six lags will be used.

3.1.2. Rolling window regression

We next examine the dynamics of the relation between bitcoin exchange reserves and bitcoin returns by estimating the regres-

Table 3

Bias Corrections.

This table reports coefficient estimates of $\Delta Reserve$ in regression model (1) with standard errors are corrected using different approaches. $ret_{t+k,t+h}$ is bitcoin returns from day $t+k$ to day $t+h$. *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

	Dependent variable:		
	ret_{t+1} (1)	$ret_{t+1,t+7}$ (2)	$ret_{t+1,t+30}$ (3)
Hodrick (1992)	-0.374* (-1.792)	-0.949 (-1.583)	-1.381 (-0.932)
Boudoukh et al. (2019)		-0.949* (-1.791)	-1.381 (-0.934)
Stambaugh (1999)	-0.297* (-1.701)		

sion model (1) using rolling two-year windows. Consistent with the baseline results in Table 2 for the full sample, Fig. 3 shows negative coefficients of $\Delta Reserve$ for all return windows over time. The Figure also reveals that the negative impact of exchange reserves on short-term returns (ret_t and ret_{t+1}) is more prominent for the first half of the sample, while the impact on long-term returns (from $ret_{t+1,t+7}$ to $ret_{t+1,t+30}$) is more significant for the second half.

Interestingly, the turning point seems to coincide with the trend of bitcoin exchange reserves which started decreasing from late 2019, signalling the possibility that there is a contemporaneous or short-term impact for transfers on exchanges (i.e., increased exchange reserves) and a long-term effect for transfers off exchanges (i.e., decreased exchange reserves).

3.1.3. Increased versus decreased exchange reserves

In this section, we test whether transfers on and off exchanges have different effects on short-term and long-term bitcoin returns. To examine this notion, we follow Chordia et al. (2002) to split $\Delta Reserve$ into positive and negative parts and include them as separate regressors, allowing us to observe a differential impact of decreased and increased exchange reserves. The model specification is:

$$ret_{t+k,t+h} = \alpha + \beta_1 \Delta Reserve_t I_{\Delta Reserve_t > 0} + \beta_2 \Delta Reserve_t I_{\Delta Reserve_t < 0} + \sum \gamma_i Control_{i,t} + \varepsilon_{t+k,t+h} \quad (2)$$

where $I_{\Delta Reserve_t > 0}$ is a dummy variable that equals one if $\Delta Reserve_t$ is positive and zero otherwise, and $I_{\Delta Reserve_t < 0}$ is a dummy variable that equals one if $\Delta Reserve_t$ is negative and zero otherwise.

The results presented in Table 4 show that the negative coefficient estimates for exchange reserve changes across all horizons reported for the baseline regression model hide a difference between positive and negative reserve changes that is uncovered with the augmented model. The coefficient estimate for positive reserve changes (β_1) remains negative and significant over short horizons but becomes insignificant over longer horizons (7 days or more). The short-term impact on returns implies that investors transferring bitcoin onto exchanges sell them shortly afterwards, creating an immediate negative price reaction. Meanwhile, the significant effect of negative exchange reserve changes on long-term returns (but not on short-term returns) is in line with our premise on the long-term holding function of private wallets and the short-term trading function of exchange accounts.

3.2. Effects of bitcoin exchange reserve changes on bitcoin volatility

We next examine the effects of bitcoin exchange reserves on bitcoin volatility. Prior studies on traditional markets show that absolute order imbalance positively affects stock or exchange rate

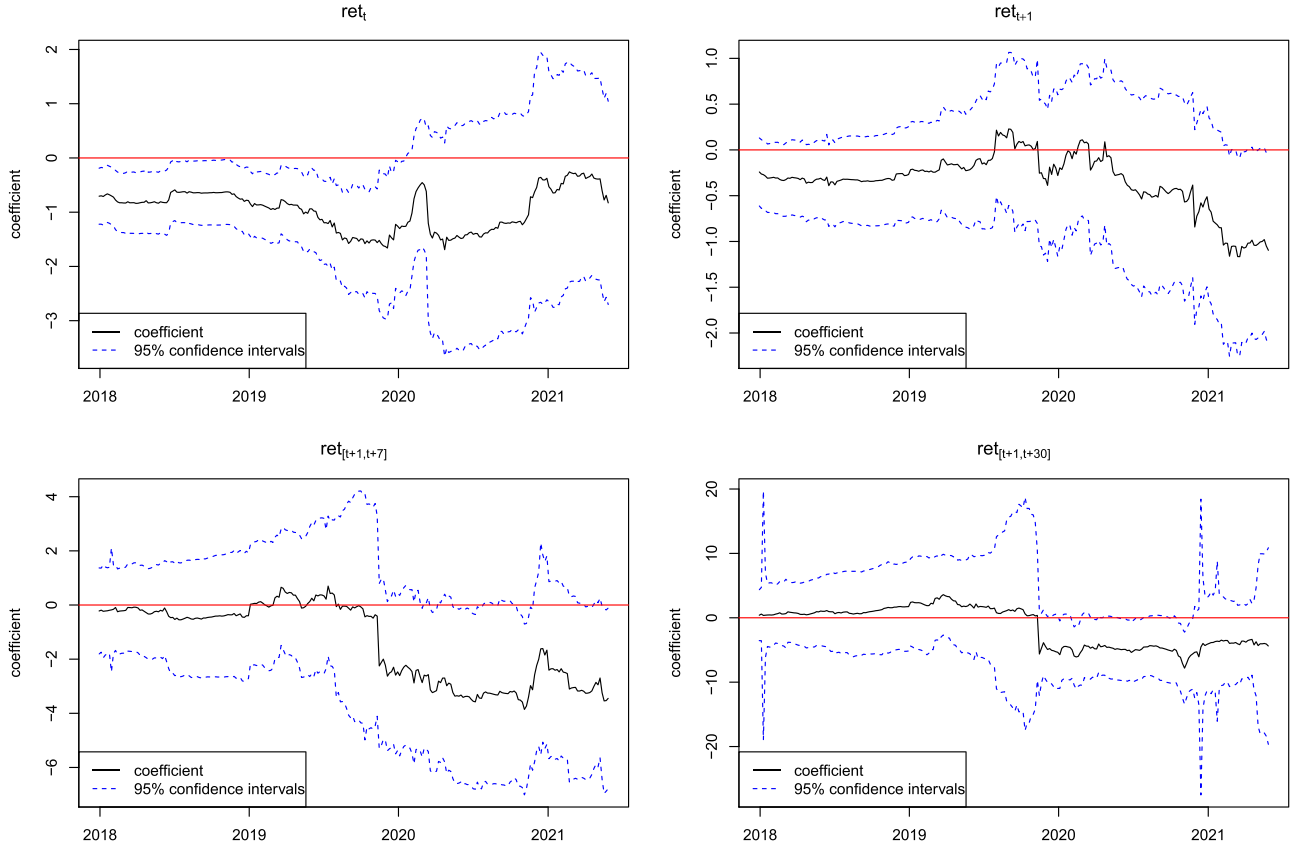


Fig. 3. Two-year rolling window regression coefficients.

This figure displays the β estimates of Eq. (1): $ret_{[t+k,t+h]} = \alpha + \beta \Delta Reserve_t + \sum \gamma_i Control_{it} + \varepsilon_t$ using two-year rolling windows. $ret_{[t+k,t+h]}$ denotes bitcoin returns from day $t+k$ to day $t+h$. $\Delta Reserve$ denotes log changes of daily bitcoin exchange reserves. The 95% confidence intervals (calculated based on Newey-West standard errors) are represented by the blue dashed lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

volatility (e.g., Berger et al., 2009; Chan and Fong, 2006; Chordia et al., 2002; Opschoor et al., 2014). Since reserve changes is used in this study as an alternative proxy for investors' intention to buy or sell instead of order imbalance, we also expect that the absolute reserve changes have a positive relation with bitcoin volatility.¹⁸ To test this notion, we augment the standard Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) proposed by Corsi (2009) with absolute reserve changes. The model is specified as follows:

$$RV_{[t+k,t+h]} = \alpha + \beta |\Delta Reserve_t| + \gamma_1 RV_{[t-1]} + \gamma_2 RV_{[t-7,t-1]} + \gamma_3 RV_{[t-30,t-1]} + \varepsilon_{[t+k,t+h]} \quad (3)$$

where $RV_{[t+k,t+h]}$ is realized volatility over the period from day $t+k$ to day $t+h$ which is calculated as the squared root of the sum of squared 5-minute returns over the period (Andersen et al., 2001), and $|\Delta Reserve_t|$ is the absolute value of $\Delta Reserve_t$.

Analogous to the return analysis, we also examine the differential effects conditional on increased and decreased reserves using the following model:

$$RV_{[t+k,t+h]} = \alpha + \beta_1 |\Delta Reserve_t| I_{\Delta Reserve_t > 0} + \beta_2 |\Delta Reserve_t| I_{\Delta Reserve_t < 0} + \gamma_1 RV_{[t-1]} + \gamma_2 RV_{[t-7,t-1]} + \gamma_3 RV_{[t-30,t-1]} + \varepsilon_{[t+k,t+h]} \quad (4)$$

¹⁸ As order imbalance is calculated as the difference between buyer-initiated trades and seller-initiated trades, an increase in order imbalance implies buying pressure and a decrease in order imbalance implies selling pressure. Analogously, in the context of exchange reserves, an increase in exchange reserves implies increased selling pressure and a decrease in exchange reserves implies decreased selling pressure. However, as we are interested in the absolute values, order imbalance and exchange reserve changes are expected to have similar effects on volatility.

The regression estimates of Eq. (3) in Table 5 show that the absolute change of bitcoin exchange reserves is positively related to bitcoin volatility contemporaneously and over one-day ahead as all β estimates are positive and statistically significant at 1% and 5% levels. However, bitcoin exchange reserves do not predict long-term (i.e., one-week and one-month-ahead) volatility, which is consistent with the overall insignificant effect of $\Delta Reserve$ on long-horizon bitcoin returns in Table 2.

Further analysis on increased and decreased reserves show that there is an asymmetric effect on future bitcoin volatility over short horizons but not over longer horizons (30 days). This is shown by the significant difference between β_1 and β_2 in columns (1)–(3) of Table 6 and the insignificant difference in column (4). The asymmetric effect indicates that positive changes of exchange reserves have a stronger impact on short-term bitcoin volatility than negative changes. As illustrated in Appendix B, this result is consistent with our previous results regarding bitcoin returns in Table 4 and with a “classical” asymmetric volatility or leverage effect generally observed in equity markets where negative returns increase the volatility by more than positive returns (Glosten et al., 1993).

3.3. Reserve change triggers

So far we have treated reserve changes as an exogenous variable and examined how they are related to contemporaneous and future returns and volatility of bitcoin. However, it is also possible that price movements affect investors' trading behavior and thus the movement of bitcoin between their private wallets and exchange accounts. Such effects have been documented in stock markets. For example, Chordia et al. (2002) show that investors in the US stock market tend to buy more following market crashes.

Table 4

Increased versus decreased reserves and bitcoin returns.

This table reports regression estimates of bitcoin concurrent and future returns on positive and negative changes of daily bitcoin reserves ($\Delta Reserve_t$). $I_{\Delta Reserve_t > 0}$ is a dummy variable that equals one if $\Delta Reserve_t$ is positive, and zero otherwise. $ret_{[t+k,t+h]}$ is bitcoin returns from day $t+k$ to day $t+h$. $\log Volume_t$ is the natural log of the daily bitcoin trading volume on exchanges in US dollars. $\log MktCap_t$ is the natural log of daily bitcoin market capitalization in US dollars. t -statistics in column (1) are adjusted using Newey-West standard errors with six lags, and in columns (2)–(4) are adjusted using Hodrick (1992) standard errors. *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

	Dependent variable:			
	ret_t (1)	ret_{t+1} (2)	$ret_{[t+1,t+7]}$ (3)	$ret_{[t+1,t+30]}$ (4)
$\Delta Reserve_t \times I_{\Delta Reserve_t > 0}$	−1.299*** (−3.364)	−0.590** (−2.017)	0.606 (0.562)	3.668 (1.219)
$\Delta Reserve_t \times I_{\Delta Reserve_t < 0}$	−0.068 (−0.213)	−0.246 (−0.706)	−3.572*** (−3.394)	−9.891*** (−3.071)
ret_t		−0.040 (−1.185)	0.057 (0.668)	0.143 (0.837)
$ret_{[t-21,t-1]}$	0.012** (2.168)	0.012* (1.874)	0.046 (1.081)	0.092 (0.687)
$\log Volume_t$	−0.002 (−1.601)	0.001 (1.144)	0.008 (0.877)	0.027 (0.766)
$\log MktCap_t$	0.003 (1.330)	−0.003 (−1.354)	−0.016 (−1.032)	−0.059 (−0.967)
Constant	−0.027 (−0.853)	0.046 (1.502)	0.242 (1.145)	0.947 (1.148)
Observations	1977	1976	1970	1947
R ²	0.019	0.008	0.019	0.034
Adjusted R ²	0.017	0.005	0.016	0.031

Table 5

Effects of bitcoin exchange reserves on volatility.

This table reports regression estimates of bitcoin concurrent and future realized volatility on the absolute value of log change of daily bitcoin reserves ($|\Delta Reserve_t|$) and control variables. $RV_{[t+k,t+h]}$ is bitcoin realized from day $t+k$ to day $t+h$, which is computed as the squared root of the sum of 5-minute squared returns over the period. t -statistics in column (1) are adjusted using Newey-West standard errors with six lags, and in columns (2)–(4) are adjusted using Hodrick (1992) standard errors. *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

	Dependent variable:			
	RV_t (1)	RV_{t+1} (2)	$RV_{[t+1,t+7]}$ (3)	$RV_{[t+1,t+30]}$ (4)
$ \Delta Reserve_t $	0.822*** (2.598)	0.656** (2.478)	1.002 (1.607)	1.425 (0.895)
RV_{t-1}	1.572*** (5.278)	0.589 (1.492)	0.757 (0.890)	0.823 (0.546)
$RV_{[t-7,t-1]}$	0.157*** (5.052)	0.160*** (5.918)	0.366*** (2.758)	0.558* (1.806)
$RV_{[t-30,t-1]}$	0.046*** (3.572)	0.053*** (4.011)	0.177** (2.181)	0.337 (1.299)
Constant	0.003*** (2.705)	0.003*** (2.661)	0.015** (2.038)	0.060* (1.937)
Observations	1977	1976	1970	1947
R ²	0.498	0.408	0.449	0.391
Adjusted R ²	0.497	0.406	0.447	0.390

To examine this possibility, we estimate the following models to examine effects of bitcoin returns and volatility on future bitcoin exchange reserves in normal market conditions (Eq. (5)) and in extreme market conditions (Eq. (6)):

$$\Delta Reserve_{t+k} = \alpha + \beta X_t + \sum \gamma_i Control_{i,t} + \epsilon_{t+k} \quad (5)$$

$$\Delta Reserve_{t+k} = \alpha + \beta_1 X_t I_{q5 \leq X_t \leq q95} + \beta_2 X_t I_{X_t < q5} + \beta_3 X_t I_{X_t > q95} + \sum \gamma_i Control_{i,t} + \epsilon_{t+k} \quad (6)$$

where X_t is either daily returns (ret_t) or daily realized volatility (RV_t) of bitcoin on day t . $I_{X_t < q5}$ ($I_{X_t > q95}$) is a dummy variable

Table 6

Increased versus decreased reserves and bitcoin volatility.

This table reports regression estimates of bitcoin concurrent and future realized volatility on the absolute value of log change of daily bitcoin reserves ($|\Delta Reserve_t|$) and control variables. $I_{\Delta Reserve_t > 0}$ is a dummy variable that equals one if $\Delta Reserve_t$ is positive, and zero otherwise. $RV_{[t+k,t+h]}$ is bitcoin realized from day $t+k$ to day $t+h$, which is computed as the squared root of the sum of 5-minute squared returns over the period. t -statistics in column (1) are adjusted using Newey-West standard errors with six lags, and in columns (2)–(4) are adjusted using Hodrick (1992) standard errors. *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

	Dependent variable:			
	RV_t (1)	RV_{t+1} (2)	$RV_{[t+1,t+7]}$ (3)	$RV_{[t+1,t+30]}$ (4)
$ \Delta Reserve_t \times I_{\Delta Reserve_t > 0} (\beta_1)$	1.155*** (3.055)	0.922*** (2.777)	1.505** (2.102)	1.792 (1.122)
$ \Delta Reserve_t \times I_{\Delta Reserve_t < 0} (\beta_2)$	0.011 (0.068)	0.008 (0.038)	−0.224 (−0.324)	0.530 (0.270)
RV_{t-1}	1.564*** (5.308)	0.586 (1.487)	0.748 (0.880)	0.821 (0.545)
$RV_{[t-7,t-1]}$	0.159*** (5.349)	0.161*** (5.974)	0.369*** (2.777)	0.560* (1.810)
$RV_{[t-30,t-1]}$	0.044*** (3.592)	0.052*** (3.917)	0.175** (2.153)	0.335 (1.294)
Constant	0.003*** (3.583)	0.004*** (3.109)	0.016** (2.154)	0.061** (1.966)
F-statistics ($H_0: \beta_1 = \beta_2$)	12.316***	8.542**	5.476*	0.834
Observations	1977	1976	1970	1947
R ²	0.515	0.418	0.456	0.393
Adjusted R ²	0.513	0.417	0.454	0.391

that equals one if X_t is smaller (larger) than its 5% (95%) quantile. $\Delta Reserve_{t+k}$ represent the exchange reserve change one day ($k = 1$) or two days ahead ($k = 2$). We control for, but omit to report for brevity, the daily bitcoin exchange reserve changes ($\Delta Reserve_t$), the natural log of bitcoin trading volume ($\log Volume_t$), and the natural log of bitcoin market capitalization ($\log MktCap_t$).

The results on the effects of bitcoin returns on bitcoin exchange reserves are presented in Table 7 Panel A. Column (1) and (3) show that on average returns have insignificant effects on the future changes of exchange reserves. Meanwhile, the negative and significant coefficients of $ret_t \times I_{ret_t < q5}$ in column (2) where $\Delta Reserve_{t+1}$ is the dependent variable indicates that large negative bitcoin returns are followed by an increase of bitcoin exchange reserves the next day. The effect disappears for longer lags and for normal and extremely good market conditions. There are several alternative mechanisms explaining these results. First, the results may be driven by panic selling behavior of bitcoin investors, i.e., they load more bitcoins onto exchange accounts after observing extremely bad market conditions with the intention to sell. Second, the results are also consistent with a stop-loss strategy where investors sell after prices drop below a certain threshold. Third, the results may stem from high leverage and margin trading offered by bitcoin exchanges. After a significant price drop, highly leveraged investors need to deposit more bitcoin onto their exchange accounts to avoid the liquidation of their positions.¹⁹

Table 7 Panel B displays results for volatility. Positive and statistical significant coefficients of RV_t and $RV_t \times I_{q5 \leq ret_t \leq q95}$ in all regressions indicate that during normal volatility periods, an increase of volatility is followed by an increase of reserve changes, implying that investors load more bitcoin from their private wallets onto exchange accounts. However, when market volatility is extremely low (i.e. daily realized volatility is smaller than its 5th percentile) or extremely high (i.e. daily realized volatility is larger than its 95th percentile), the relation between RV and subsequent $\Delta Reserve$ is

¹⁹ While all three mechanisms are theoretically plausible, an empirical analysis on what mechanism plays a dominant role is beyond the scope of this paper, and could be an avenue for future research.

Table 7

Effects of bitcoin returns and volatility on bitcoin exchange reserves. This table reports regression estimates of daily log change of bitcoin exchange reserves ($\Delta Reserve_{t+k}$) on lagged bitcoin returns (ret_t) and realized volatility (RV_t). $I_{ret_t < q5}$ ($I_{ret_t > q95}$) is a dummy variable that equals one if ret_t is smaller (larger) than its 5th (95th) percentile and zero otherwise. $I_{RV_t < q5}$ ($I_{RV_t > q95}$) is a dummy variable that equals one if RV_t is smaller (larger) than its 5th (95th) percentile and zero otherwise. Control variables are included but not reported. t -statistics are adjusted using Newey-West standard errors with six lags and reported in parentheses. *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

	Dependent variable:			
	$\Delta Reserve_{t+1}$		$\Delta Reserve_{t+2}$	
	(1)	(2)	(3)	(4)
Panel A: Effects of bitcoin returns				
ret_t	-0.002 (-0.447)		-0.001 (-0.246)	
$ret_t \times I_{q5 \leq ret_t \leq q95}$		0.003 (0.450)		-0.007 (-1.257)
$ret_t \times I_{ret_t < q5}$		-0.011** (-2.346)		0.001 (0.149)
$ret_t \times I_{ret_t > q95}$		0.005 (0.850)		0.003 (0.577)
Control variables	Yes	Yes	Yes	Yes
Observations	1977	1977	1977	1977
R^2	0.010	0.013	0.007	0.008
Adjusted R^2	0.008	0.010	0.005	0.005
Panel A: Effects of bitcoin volatility				
RV_t	0.011* (1.755)		0.015* (1.849)	
$RV_t \times I_{q5 \leq RV_t \leq q95}$		0.023* (1.952)		0.029** (2.421)
$RV_t \times I_{RV_t < q5}$		-0.296* (-1.654)		-0.107 (-0.601)
$RV_t \times I_{RV_t > q95}$		0.008 (1.171)		0.012 (1.416)
Control variables	Yes	Yes	Yes	Yes
Observations	1977	1977	1977	1977
R^2	0.011	0.015	0.010	0.012
Adjusted R^2	0.009	0.012	0.008	0.009

insignificant. The findings are consistent with the “loaded for bear” analogy, i.e. increased risk triggers the loading of the gun but only if the level of risk is not too low or already too high.

4. Robustness tests

4.1. Out-of-sample forecast

In this section, we test whether $\Delta Reserve$ contributes to forecasting accuracy of future bitcoin returns and volatility in an out-of-sample setting. Guided by the in-sample results which show that $\Delta Reserve$ has a significant effect on the one-day-ahead returns, we examine model (1) with ret_{t+1} as the dependent variable and examine whether it has better forecasting accuracy on the next-day returns compared to the benchmark model without $\Delta Reserve$. Similarly, we examine the forecasting power of $|\Delta Reserve|$ on the next-day volatility RV_{t+1} using model (3). For both returns and volatility, we also consider models without control variables, i.e., only the intercepts are included in the benchmark models.

We use either one-, two-, or three-year rolling windows as training samples to generate daily forecast values of returns and volatility. To compare the forecasting models, we first calculate the mean squared errors (MSE) of each model as

$$MSE = \frac{1}{T-p} \sum_{t=p}^{T-1} (y_{t+1} - \hat{y}_{t+1})^2 \quad (7)$$

where y_{t+1} and $\hat{y}_{t+1}$ are observed and forecast values of the variable of interest (either returns or volatility), T is the number of observations in the full sample, and p is the number of observations in the rolling samples. We then use two pairwise tests including R_{OS}^2 proposed by Campbell and Thompson (2008) and Welch and Goyal (2008) and z-statistics proposed by Clark and West (2007) to compare full and nested predictive models.

The Campbell and Thompson (2008) and Welch and Goyal (2008) forecast evaluation is calculated as

$$R_{OS}^2 = 1 - \frac{MSE_u}{MSE_r} \quad (8)$$

where u and r respectively denotes the unrestricted model and the restricted model without $\Delta Reserve$. A positive R_{OS}^2 is interpreted as $\Delta Reserve$ improving the forecasting accuracy of the benchmark model and vice versa.

Clark and West (2007) on the other hand rely on the statistical significance of the test statistic z specified as:

$$z = \sqrt{MSE_r} - (\sqrt{MSE_u} - \kappa_u) \quad (9)$$

where $\kappa_u = \frac{1}{T-p} \sum_{t=p}^{T-1} (\hat{y}_{u,t+1} - \hat{y}_{r,t+1})$. Under the null hypothesis that $\Delta Reserve$ does not improve the forecasting accuracy, the critical values at 10% and 5% levels are 1.282 and 1.645 respectively.

Table 8 shows that in general values of R_{OS}^2 are positive and the z-statistics are statistically significant in all out-of-sample forecasting designs except for the return forecast using one-year training windows. The results indicate that the $\Delta Reserve$ -augmented models outperform the benchmark models in forecasting the one-day-ahead returns and volatility, consistent with the in-sample evidence on the predictive power of bitcoin exchange reserves on future bitcoin prices.

We further follow Campbell and Thompson (2008) to assess the economic significance of the improvement in the return forecasts by comparing R_{OS}^2 to the unconditional Sharpe ratio of bitcoin (SR).²⁰ Assume that a mean-variance investor holds a portfolio of bitcoin and a risk-free asset, the proportional increase in the expected portfolio return is approximately equal to the ratio of $\frac{R_{OS}^2}{SR^2}$. Given that the Sharpe ratio of daily bitcoin returns in our sample is 0.054, the R_{OS}^2 of 0.285% using two-year rolling windows in the model with control variables results in the ratio of 0.977, implying that the forecast can increase the average daily portfolio returns by 97.7%. This corresponds to an actual increase in the expected returns of $(\frac{1}{\lambda})(\frac{R_{OS}^2}{1-R_{OS}^2})(1+SR^2)$ where λ is the investor's degree of relative risk aversion. For example, a λ of five as used in Neely et al. (2014) results in an increase of nearly 0.06% per day or 20.9% per year. The economic significance of the forecast is even larger in the return model with three-year rolling windows due to a higher value of R_{OS}^2 . Overall, the out-of-sample tests show that the inclusion of $\Delta Reserve$ improves the power of the forecasting models both economically and statistically.

4.2. Hourly frequency returns

We further examine the relation between bitcoin exchange reserve changes and bitcoin returns using more granular data at an hourly frequency. Consistent with our hypothesis on selling pressure the regression results in Table 9 show that the coefficients of $\Delta Reserve_h$ are negative in regressions with both concurrent and future returns in the next one, two, and three hours. However, $\Delta Reserve_h$ only exhibits a statistically significant effect for the next hour returns, implying that significant selling is more likely to be

²⁰ As proposed in Campbell and Thompson (2008), the assessment is only relevant to the return forecast model.

Table 8

Out-of-sample forecast of bitcoin exchange reserves on bitcoin prices.

This table presents Campbell and Thompson's (2008) R_{OS}^2 and Clark and West's (2007) z-statistics using one-year, two-year, and three-year training windows. Panel A and Panel B respectively present the results of models without and with control variables. A positive R_{OS}^2 or a statistically significant z-statistics is interpreted as $\Delta Reserve$ improving the forecasting accuracy of the benchmark model. * and ** denote statistical significance at 10% and 5% respectively.

Window	Criteria	Panel A: Without controls		Panel B: With controls	
		Returns	Volatility	Returns	Volatility
One year	R_{OS}^2 (%)	−0.021	6.411	−0.051	1.506
	z-statistics	1.042	5.502**	1.156	3.279**
Two years	R_{OS}^2 (%)	0.211	6.936	0.285	1.805
	z-statistics	1.575*	8.329**	1.867**	2.484**
Three years	R_{OS}^2 (%)	0.312	8.521	0.405	1.679
	z-statistics	2.059**	6.694**	2.444**	2.720**

Table 9

Effects of bitcoin exchange reserve changes on bitcoin returns, hourly frequency.

This table reports the estimated effects of the hourly bitcoin reserve changes ($\Delta Reserve_h$) on bitcoin concurrent and future hourly returns. $\log MktCap$ is the natural log of hourly bitcoin market capitalization in US dollars. t-statistics are adjusted using Newey-West standard errors and reported in parentheses. *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

	Dependent variable:			
	ret_h (1)	ret_{h+1} (2)	ret_{h+2} (3)	ret_{h+3} (4)
$\Delta Reserve_h$	−0.015 (−0.289)	−0.090** (−2.328)	0.001 (0.011)	−0.001 (−0.019)
ret_h		0.016* (1.783)	−0.027*** (−3.140)	0.008 (1.002)
ret_{h-1}	0.015* (1.768)	−0.027*** (−3.165)	0.008 (1.021)	−0.014 (−1.403)
$\log MktCap_h$	0.00001 (0.377)	−0.00003 (−0.997)	−0.00003 (−0.999)	−0.00002 (−0.977)
Constant	−0.0001 (−0.231)	0.001 (1.171)	0.001 (1.169)	0.001 (1.146)
Observations	47,479	47,479	47,479	47,479
R^2	0.0002	0.001	0.001	0.0003
Adjusted R^2	0.0002	0.001	0.001	0.0002

observed one hour after investors transfer bitcoin to exchange wallets.

4.3. Individual exchanges

In this section, we examine the effects of bitcoin reserves of individual exchanges on bitcoin returns and volatility. Figure 4 shows the proportions of reserves of individual exchanges to total exchange reserves. Among 20 exchanges, Coinbase holds the largest amount of reserves at all times, which accounts for around 25% of total exchange reserves by the end of the sample period. Binance, although founded later than others, exhibits a significant increase of reserves in recent years. Meanwhile, other well known exchanges such as Bitfinex, BitMex or Poloniex exhibit decreasing reserves over time.

Figure 5 presents the β estimates of model (1) for four return horizons, in which $\Delta Reserve$ is calculated separately for each individual exchange. In line with the overall negative effect on returns of the aggregate reserves, the results show that the coefficients of individual exchange reserves are negative across almost all exchanges and horizons. However, none of them shows a consistent statistical significance across horizons, indicating that the aggregate exchange reserves contain more information than that of any individual exchange in explaining and predicting future bitcoin returns and volatility.

Figure 6 presents the estimates for bitcoin volatility (Eq. (3)) and shows significant estimates of individual exchange reserve changes on bitcoin volatility, especially for big exchanges such as Coinbase, Bitfinex or Kraken. The results are consistent across horizons, which is intuitive given the overlaps of horizons and the persistence of volatility. However, similar to the case for returns, the magnitude of individual exchange reserve changes on volatility is smaller than that of the aggregate amount, shown by significant smaller coefficient estimates.

We next relate the β estimates in Figs. 5 and 6 to the levels of bitcoin reserves of individual exchanges. The results are presented in Fig. 7a for returns and in Fig. 7b for volatility. The y-axes represent the average of individual exchanges' bitcoin reserves over the sample period. The graphs show that the effects of reserve changes on contemporaneous and future bitcoin returns and volatility are stronger among exchanges that hold a larger fraction of total reserves.

4.4. Price pressure as the mechanism

If price pressure is the underlying mechanism explaining the negative relation between exchange reserves and bitcoin prices, we should see a negative relation between exchange reserve changes and order imbalance which is calculated as the difference between buyer-initiated and seller-initiated trading volumes. In other words, investors who load bitcoin from their private wallets to exchange accounts sell the bitcoin on an exchange. However, as order imbalance is difficult to obtain in the bitcoin market as discussed earlier in Section 1, we are unable to provide comprehensive direct evidence supporting this conjecture. Instead, we use two approaches to address this concern. First, we examine the relation between exchange reserves and order imbalance of three particular exchanges - Binance, Kraken, and Bitfinex - for which we are able to collect comprehensive data on historical signed trades which are used to compute order imbalance. Second, we rely on indirect evidence by examining whether on-chain activities between private wallets and exchange accounts eventually result in off-chain on-exchange trading activities. We expect to observe a positive relation between the total bitcoin trading volume (sum of buy and sell volumes) on exchanges and the total bitcoin transferred in and out exchanges' addresses on blockchain (exchange-related on-chain transactions). Both approaches yield results consistent with the price pressure hypothesis.

4.4.1. Exchanges' order imbalance

We collect historical trades of all trading pairs on Binance, Kraken, and Bitfinex in which bitcoin is either the base asset or the quote asset through the exchanges' Application Programming Interface (API). The data are available from July 14, 2017 for Bi-

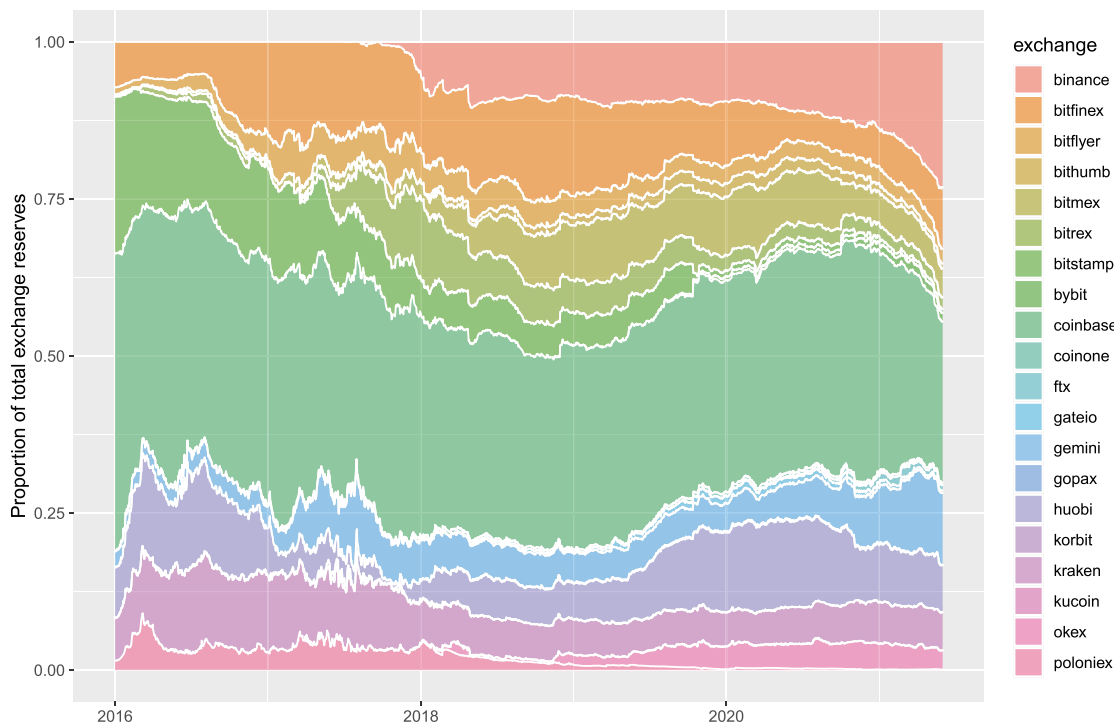


Fig. 4. Bitcoin reserve shares by individual exchanges.

nance, and from January 01, 2016 for Kraken and Bitfinex. For pairs in which bitcoin is the quote asset, trades that are labelled buyer-initiated by the exchanges are considered seller-initiated with regard to bitcoin, and vice versa.²¹ Aggregating trades in all pairs for each exchange, we then compute the bitcoin order imbalance variable $OIB_BTC_{i,t}$ as the number of buyer-initiated bitcoins purchased less the number of seller-initiated bitcoins sold on day t on exchange i .²² Finally, we regress the daily order imbalance measure on daily changes of the corresponding exchange's bitcoin reserves and control variables.²³

The results presented in Table 10 Panel A show that the daily reserve changes are negatively related to the contemporaneous and next-day order imbalance. Further analysis that separates increased and decreased reserves shows that the negative effect is attributed to the former, consistent with our premise that investors transferring bitcoin onto exchanges will sell bitcoin. On the other hand, insignificant coefficients of decreased reserves are in line with the fact that transfers off exchanges to private wallets are not necessarily associated with more purchases.

We note that while exchange reserve changes and order imbalance are two alternative proxies for price pressure, they are not exactly similar. For example, R^2 in the regression of order imbalance on daily changes of exchange reserves for Binance (column (1) of Table 10) is only 17%. The discrepancies are intuitive and thus expected because of at least two reasons. First, investors who purchase bitcoins on exchanges (thus leading to an increase in OIB_BTC) may choose to keep them in their exchange accounts rather than transfer to their private wallets, rendering $\Delta Reserve$ unchanged. Second, investors who load bitcoins onto exchanges

may not sell them on the same day but can wait until the next day or the next few days, leading to an imperfect contemporaneous co-movement between $\Delta Reserve$ and OIB_BTC . This is evidenced by the significant negative relation between positive $\Delta Reserve$ on day t and OIB_BTC the next day $t + 1$ shown in columns (2) and (4) of Table 10.

We next examine the effects of bitcoin order imbalance on bitcoin returns and volatility in comparison with that of bitcoin exchange reserves. We calculate the combined order imbalance measure (OIB_BTC_{BBK}) as the difference between the number of buyer-initiated and seller-initiated bitcoins traded on all three exchanges Binance, Bitfinex and Kraken. For a fair comparison, we only use bitcoin reserves of these three exchange and calculate $\Delta Reserve_{BBK}$ as the log change of their total daily bitcoin reserves. Results in Table 11 Panel A show that although order imbalance is strongly related to returns on the same day, it has no predictive power on the next day bitcoin returns. Meanwhile, the effect of exchange reserve changes on the next day bitcoin returns is significant and robust after controlling for order imbalance. Panel B confirms the results in the out-of-sample setting where $\Delta Reserve_{BBK}$ significantly improves the forecasting power of the benchmark (i.e. intercept only) model when using two-year or three-year training periods, while the inclusion of OIB_BTC_{BBK} reduces the forecasting power. Taken together, the results imply that while order imbalance reflects current price pressure that immediately drives down the prices, exchange reserve changes reflect future price pressure which affects future returns. This further highlights the differences between exchange reserve changes and order imbalance discussed in the previous paragraph.

4.4.2. Exchange-related on-chain transactions and on-exchange trading volume

For the analysis in this section, ideally the on-exchange bitcoin trading volumes of all 20 exchanges covered in this study should be used, which unfortunately are not available to us. Hence we use two alternative proxies. First, we use data from Coinmarketcap.com which provides the aggregate bitcoin trading volume (in BTC and

²¹ For example, buying ETH/BTC is equivalent to selling BTC/ETH.

²² While Chordia et al. (2002) also use OIB_NUM which is the difference between the numbers of buyer-initiated and seller-initiated trades as an alternative measure of order imbalance, we do not consider the measure in this study because its relation with reserve changes is unclear (i.e., more bitcoins loaded onto an exchange are not necessarily associated with a higher number of selling trades and vice versa).

²³ For example, Binance's OIB_BTC is regressed on Binance's $\Delta Reserve$.

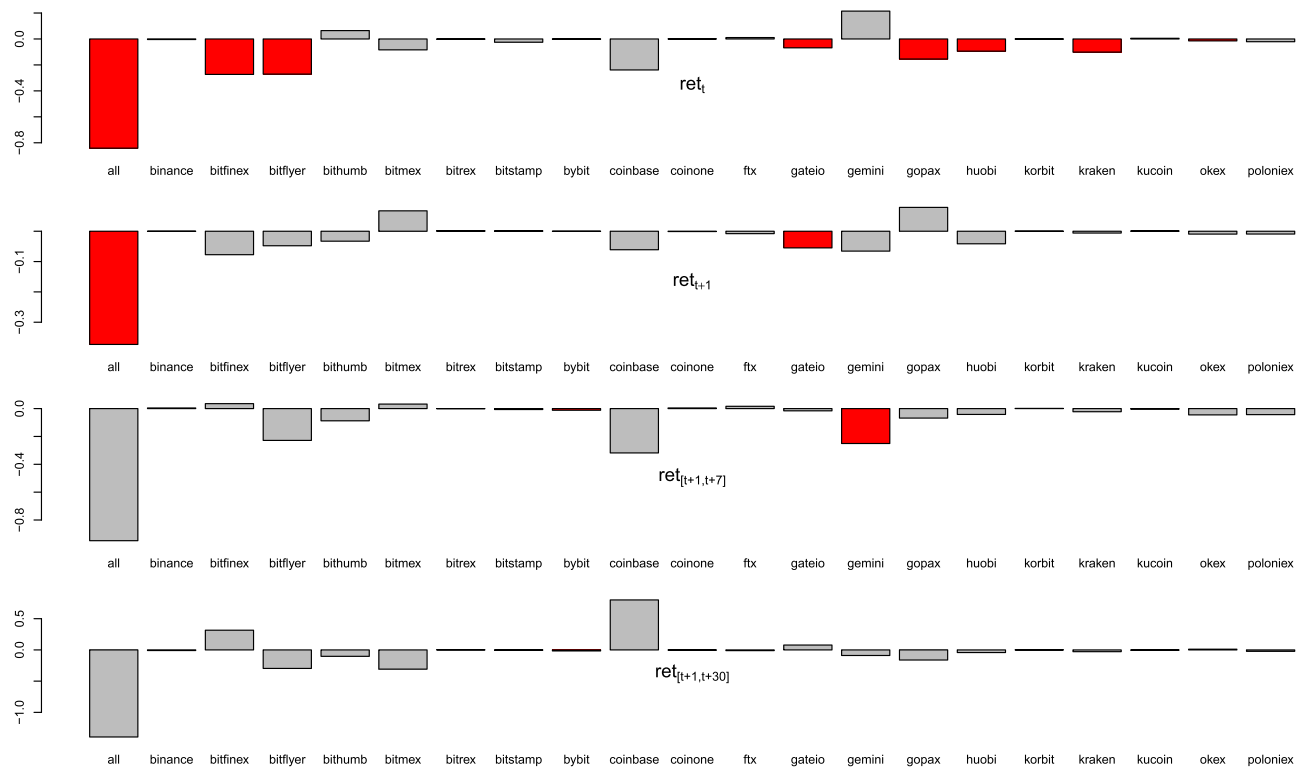


Fig. 5. Effect of individual exchange reserves on bitcoin returns.

This figure presents the beta estimates of the return model (1), in which $\Delta Reserve$ is calculated separately for each individual exchange. Standard errors of regression coefficients are calculated following Hodrick (1992). Red bars represent negative coefficients that are statistically significant at 10% level. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

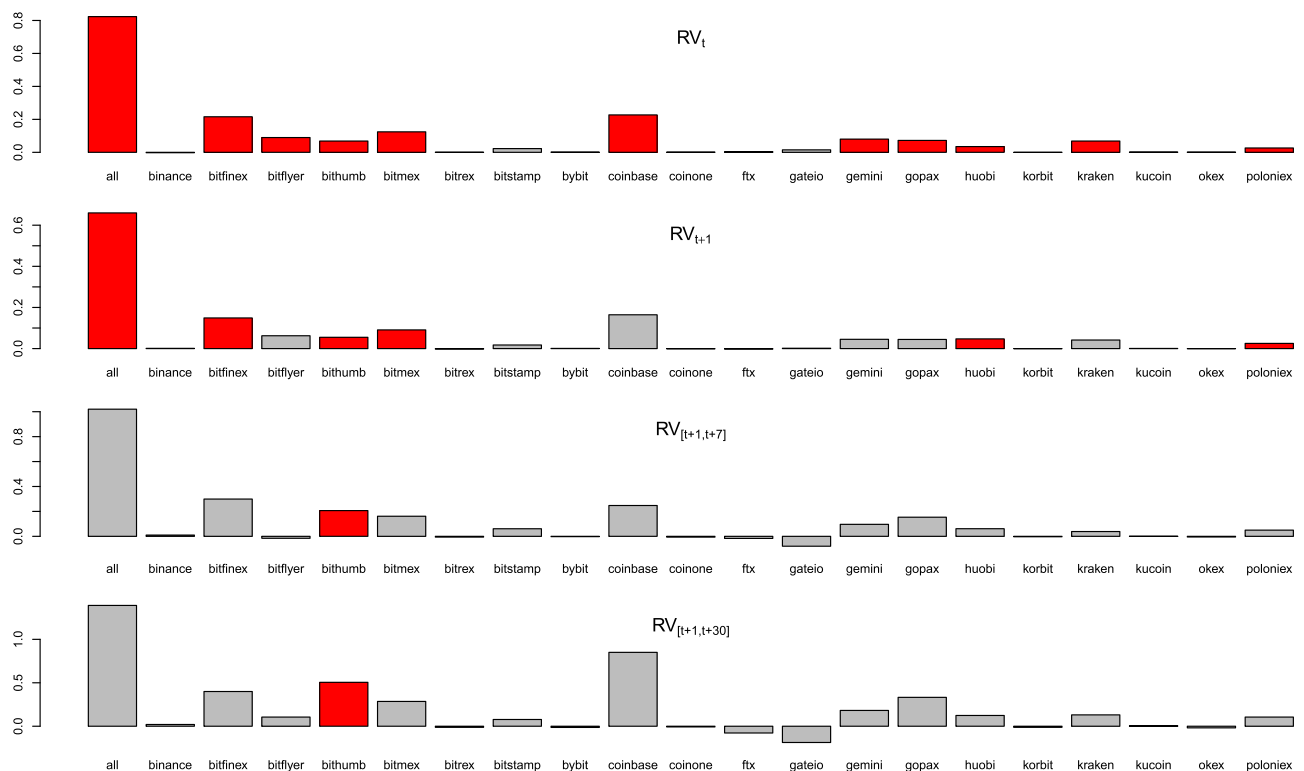


Fig. 6. Effect of individual exchange reserves on bitcoin volatility.

This figure presents the beta estimates of the volatility model (3), in which $\Delta Reserve$ is calculated separately for each individual exchange. Standard errors of regression coefficients are calculated following Hodrick (1992). Red bars represent positive coefficients that are statistically significant at 10% level. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 10

Individual exchanges' daily reserve changes and order imbalance.

This table presents the relation between bitcoin exchange reserve changes and order imbalance for three individual exchanges Binance, Kraken, and Bitfinex. $\Delta Reserve_{BN,t}$, $\Delta Reserve_{KK,t}$, and $\Delta Reserve_{BF,t}$ denote log changes of daily bitcoin reserves of Binance, Kraken, and Bitfinex respectively. OIB_BTC_t is calculated as the difference between the number of buyer-initiated and seller-initiated bitcoins traded on the exchange on day t . *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

	Dependent variable:			
	OIB_BTC_t (1)	OIB_BTC_{t+1} (2)	OIB_BTC_t (3)	OIB_BTC_{t+1} (4)
Panel A: Binance				
$\Delta Reserve_{BN,t}$	−0.084*** (−3.136)	−0.070** (−2.448)		
$\Delta Reserve_{BN,t} \times I_{\Delta Reserve_{BN,t} > 0}$			−0.084** (−2.375)	−0.075** (−1.985)
$\Delta Reserve_{BN,t} \times I_{\Delta Reserve_{BN,t} < 0}$			−0.084 (−1.372)	−0.059 (−0.890)
Control variables	Yes	Yes	Yes	Yes
Observations	1418	1417	1418	1417
R ²	0.168	0.035	0.168	0.035
Adjusted R ²	0.165	0.032	0.165	0.031
Panel B: Kraken				
$\Delta Reserve_{KK,t}$	−0.033*** (−3.619)	−0.026** (−2.045)		
$\Delta Reserve_{KK,t} \times I_{\Delta Reserve_{KK,t} > 0}$			−0.087*** (−5.803)	−0.065*** (−3.154)
$\Delta Reserve_{KK,t} \times I_{\Delta Reserve_{KK,t} < 0}$			0.012 (0.880)	0.007 (0.390)
Control variables	Yes	Yes	Yes	Yes
Observations	1978	1977	1978	1977
R ²	0.481	0.012	0.486	0.015
Adjusted R ²	0.480	0.010	0.485	0.012
Panel C: Bitfinex				
$\Delta Reserve_{BF,t}$	−0.155*** (−8.762)	−0.033 (−1.310)		
$\Delta Reserve_{BF,t} \times I_{\Delta Reserve_{BF,t} > 0}$			−0.240*** (−8.897)	−0.009 (−0.245)
$\Delta Reserve_{BF,t} \times I_{\Delta Reserve_{BF,t} < 0}$			−0.064** (−2.293)	−0.057 (−1.450)
Control variables	Yes	Yes	Yes	Yes
Observations	1971	1970	1971	1970
R ²	0.493	0.004	0.498	0.004
Adjusted R ²	0.492	0.001	0.496	0.001

Table 11

Effects of bitcoin order imbalance on bitcoin returns.

This table compares the effects of exchange reserve changes ($\Delta Reserve_{BBK}$) and order imbalance (OIB_BTC_{BBK}) of three exchanges Binance, Bitfinex, and Kraken on bitcoin returns. $\Delta Reserve_{BBK}$ is the daily log changes of total bitcoin reserves on the three exchanges. OIB_BTC_{BBK} is the difference between the number of buyer-initiated and seller-initiated bitcoins traded on the three exchanges. Panel A presents the in-sample regression results. Panel B presents the out-of-sample forecasts where we compare the benchmark intercept only model with either $\Delta Reserve_{BBK}$ -augmented model or OIB_BTC_{BBK} -augmented model. To reduce the variation in the order imbalance measure for out-of-sample forecasts, OIB_BTC_{BBK} is scaled to zero mean and unit variance (results in Panel A are qualitatively similar if scaled order imbalance is used). A positive R^2_{OS} or a statistically significant z-statistics is interpreted as the corresponding factor improving the forecasting accuracy of the benchmark model without the factor. *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

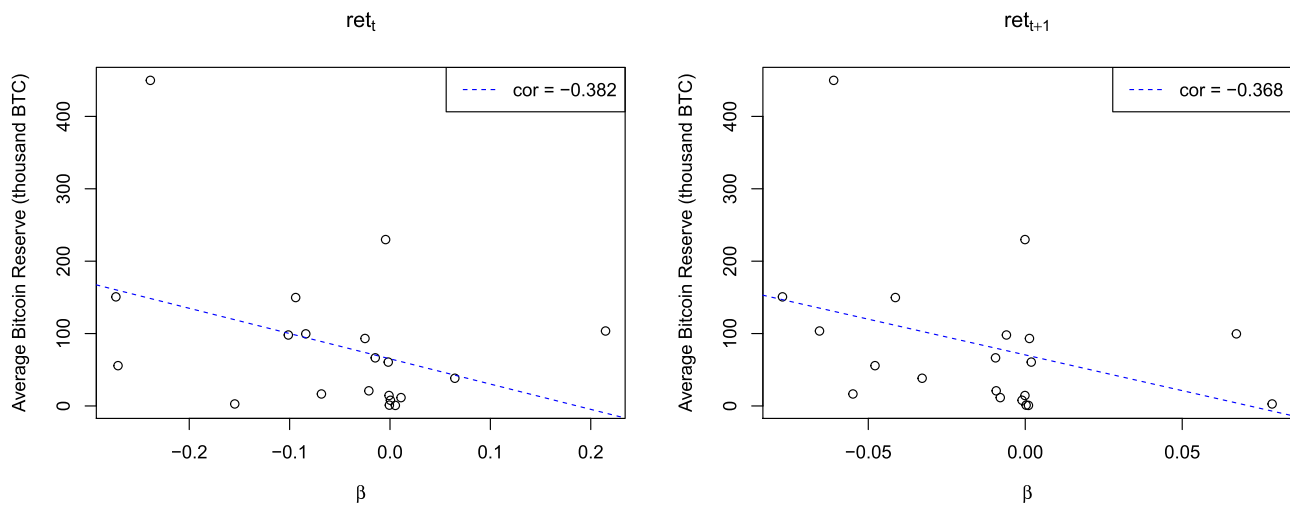
Panel A: In-sample regression

	Dependent variable:					
	ret_t			ret_{t+1}		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta Reserve_{BBK,t}$	−0.186** (−2.487)		−0.027 (−0.827)	−0.081* (−1.917)		−0.077* (−1.874)
$OIB_BTC_{BBK,t} \times 10^{-6}$		5.120*** (23.140)	5.113*** (23.273)		0.429 (1.356)	0.414 (1.305)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1977	1978	1977	1976	1977	1976
R ²	0.011	0.595	0.595	0.006	0.006	0.008
Adjusted R ²	0.009	0.594	0.594	0.003	0.004	0.005

Panel B: Out-of-sample forecast

Training Period	Predictor:			
	$\Delta Reserve_{BBK}$		OIB_BTC_{BBK} (scaled)	
	R^2_{OS} (%)	z-statistics	R^2_{OS} (%)	z-statistics
One year	−0.063	0.582	−0.668	−1.477
Two years	0.207	1.573*	−0.165	−0.144
Three years	0.233	1.356*	−0.199	−0.789

(a) Returns



(b) Volatility

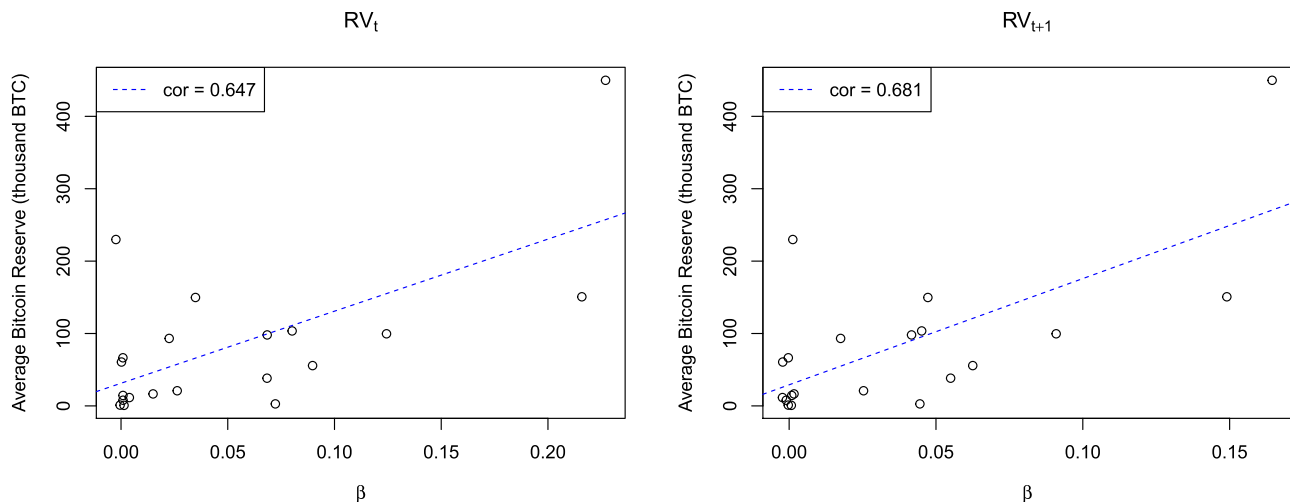


Fig. 7. Reserve levels and magnitudes of effects.

The x-axes in Panel (a) and (b) respectively represent the β estimates of Eqs. (1) and (3) regarding the effects of bitcoin exchange reserve changes on bitcoin returns and volatility for each individual exchanges. The y-axes represent the daily average of corresponding individual exchanges' bitcoin reserves.

USD) of more than 300 exchanges globally, including the 20 exchanges in our sample. Despite being a subset, those 20 exchanges are among the largest bitcoin exchanges, and thus account for a large proportion of the all-exchange aggregate trading volume provided by Coinmarketcap.com. Second, we collect the total trading volume in USD of all cryptocurrencies traded on each of the 20 exchanges under examination from Nomics.com.²⁴ Since bitcoin is the largest cryptocurrency, we postulate that it accounts for a large proportion of the all-cryptocurrency trading volumes of individual exchanges provided by Nomics.com.

Figure 8a shows that daily trading volumes on exchanges are positively correlated with daily exchange-related on-chain transactions, implying that the more frequently investors transfer their bitcoin on and off their exchange accounts, the more trading activities they conduct on exchanges. To avoid possible spurious correlations due to increased trading volumes over time, we present the

results for daily log changes in Fig. 8b, which shows even stronger correlations.

In general, the strong relation between bitcoin trading volume on exchanges and exchange-related on-chain transactions provide further support for the price pressure explanation of the negative relation between bitcoin exchange reserves and bitcoin returns.

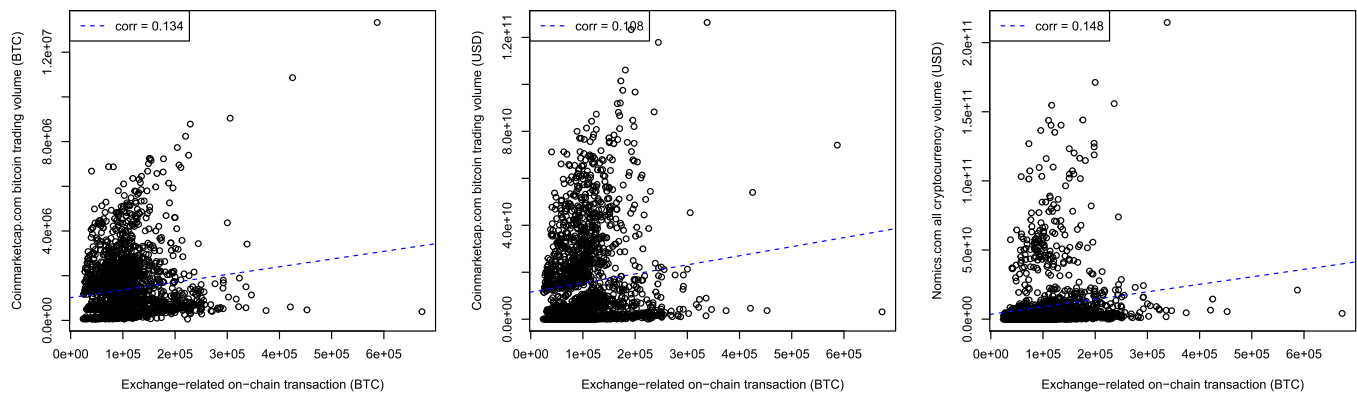
5. Comparison with other determinants of bitcoin returns

In this section, we compare the predictive power of exchange reserves on future bitcoin returns against other factors documented in prior studies which are summarized in Table 12. Most of them either focus on the contemporaneous relation and thus do not imply predictability or only examine the in-sample lead-lag effect. The only exception that conducts an out-of-sample evaluation is Bleher and Dimpfl (2019) who find an insignificant predictability of bitcoin's Google Search Volume on future bitcoin returns.

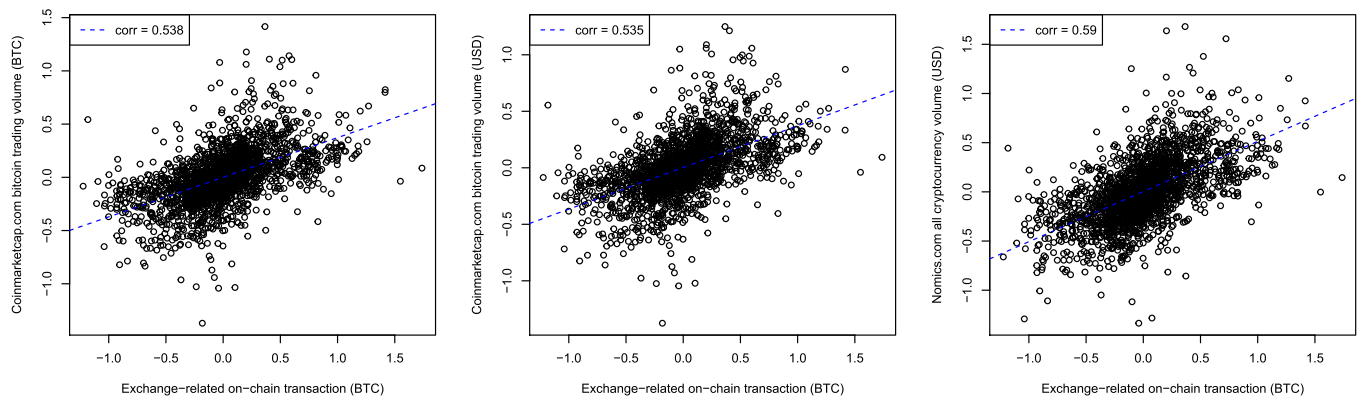
We first make a comparison in an in-sample setting by including all factors described in Table 12 as control variables in

²⁴ <https://nomics.com/exchanges>.

(a) Daily volumes



(b) Log changes of daily volumes

**Fig. 8.** Exchange-related on-chain transactions versus trading volume on exchanges.

Exchange-related on-chain transactions (x-axes) are calculated as the total of bitcoin transferred into and out of exchanges' addresses on the bitcoin blockchain. Trading volume on exchanges (y-axes) are either total bitcoin trading volumes on all exchanges collected from Coinmarketcap.com or the trading volumes of all cryptocurrencies on 20 exchanges in this study collected from Nomics.com.

Table 12
Previous Studies.

Study	Factor	Temporal Effect	Econometric Model	Out-of-sample Forecast
Market-based factors				
Aalborg et al. (2019)	Google Search Volume (SVI) Stock Volatility Index (VIX) Realized Volatility	Predictive	Linear Regression	No
Balcilar et al. (2017)	Trading Volume	Predictive	Causality-in-quantiles	No
Bleher and Dimpfl (2019)	Google Search Volume	Predictive	Vector Autoregressive Model	Yes
Bouri et al. (2019)	Trading Volume	Predictive	Granger Causality in Distribution	No
Dastgir et al. (2019)	Google Search Volume	Predictive	Granger Causality in Distribution	No
Demir et al. (2018)	Economic Policy Uncertainty	Predictive	Vector Autoregressive Model	No
Kristoufek (2015)	Google Search Volume Wikipedia Pageviews	Contemporaneous	Wavelets	No
Leirvik (2021)	Volatility of Liquidity	Contemporaneous	Linear Regression	No
Naeem et al. (2020)	Trading Volume	Contemporaneous	GARCH-copula	No
Shen et al. (2019)	Number of tweets	Predictive	Vector Autoregressive Model	No
Wang et al. (2020)	Economic Policy Uncertainty	Contemporaneous	DCC-GARCH	No
Blockchain-based factors				
Aalborg et al. (2019)	Number of unique addresses On-chain transaction volume	Predictive	Linear Regression	No
Hau et al. (2021)	Number of transaction count Number of unique addresses	Predictive	Quantile-on-quantile regression	No
Koutmos (2018)	Number of transaction count Number of unique addresses	Predictive	Vector Autoregressive Model	No
Liu and Tsyvinski (2020)	Number of wallet users Number of active addresses Number of transaction count Number of payment count	Contemporaneous	Linear Regression	No

Eq. (1). Specifically, we additionally control for bitcoin (il)liquidity using the Amihud's (2002) illiquidity measure (*Illiq*) which is computed as the ratio of absolute return to trading volume in million US dollars, and realized volatility (*RV*) which is computed as the square root of the sum of squared 5-minute returns over the day (Andersen et al., 2001). We also control for macroeconomic conditions, including ΔVIX which is computed as the daily log change of the CBOE Volatility Index and ΔEPU which is computed as the daily log change of the economic policy uncertainty index. The economic policy uncertainty index is developed by Baker et al. (2016) and can be downloaded from the authors' website (policyuncertainty.com). Since bitcoin is traded 24/7, we fill missing values of macroeconomic variables during weekends and holidays by the last prevailing values. Further control variables include volatility of liquidity (*VolaLiq*) calculated following Leirvik (2021), log change of Google Search Volume on the term "bitcoin" (ΔSVI), log change of the number of bitcoin's Wikipedia page views ($\Delta WikiSearch$), natural log of the number of tweets about bitcoin on Twitter (*logTweet*), log change of on-chain transaction volume ($\Delta TransVol$), and the bitcoin network factor (*PC_Network*). Following Liu and Tsyvinski (2020), *PC_Network* is constructed as the first principal component of four network measures, including the number of wallet users, the number of active addresses, the number of transaction count, and the number of payment count. Results in Table 13 show that almost all competing factors except momentum ($ret_{[t-21,t-1]}$) are statistically insignificant in the predictive models, while $\Delta Reserve$ remains significant after controlling for all these factors.

We further compare exchange reserves with other competing factors in an out-of-sample context by respectively replacing the variable X in the predictive model:

$$ret_{t+1} = \alpha + \beta X_t + \varepsilon_{t+1} \quad (10)$$

by $\Delta Reserve$ and each competing factor and examining whether X can improve the forecasting accuracy of the next-day bitcoin returns compared with the benchmark model. The evaluation criteria are similar to Section 4.1. The results presented in Table 14 indicate that except for $\Delta Reserve$, none of the examined factors outperforms the benchmark model as evidenced by negative R_{OS}^2 in most cases and insignificant z-statistics in all cases. Thus, exchange reserve changes survive both in-sample and out-of-sample tests as the best predictor of future returns.

6. Identification concern

The data set provided by CryptoQuant only covers 20 exchanges out of more than 300 existing exchanges, underestimating the amount of exchange reserves. However, since we do not focus on the estimation of exchange reserves levels but how changes of the reserves predict future bitcoin prices, we postulate that a sample of the largest exchanges used in this study is sufficient to capture such effect. For example, Entrop et al. (2020) shows that trading volume is a strong determinant of bitcoin price discovery, implying that prices are formed primarily on the largest exchanges. In fact, Fig. 2 Panel B shows that those 20 exchanges usually account for more than 80% of total on-chain transactions.

To further alleviate the concern, we examine the exchange reserves measures from alternative data vendors, including Glassnode, CoinMetrics, and IntoTheBlock, which respectively cover 18, 9, and 10 bitcoin exchanges. In addition, we adopt a bottom-up approach to collect addresses of 87 exchanges provided by WalletExplorer.com and calculate the exchanges reserves over time by accumulating daily bitcoin inflows into and outflows from those

Table 13

Effects of bitcoin exchange reserves on bitcoin returns with extended control variables.

This table reports regression estimates of bitcoin concurrent and future returns on log change of daily bitcoin reserves ($\Delta Reserve$) and extended set of control variables. $ret_{[t+k,t+h]}$ is bitcoin returns from day $t+k$ to day $t+h$. *logVolume* is the the natural log of the daily bitcoin trading volume on exchanges in US dollars. *logMktCap* is the natural log of daily bitcoin market capitalization in US dollars. ΔVIX is the daily log change of the CBOE Volatility Index. ΔEPU is the daily log change of the Baker et al.'s 2016 economic policy uncertainty index. ΔSVI is the daily log change of Google Search Volume on the term "bitcoin". $\Delta WikiSearch$ is the daily log change of the number of bitcoin's Wikipedia page views. $\Delta TransVol$ is the daily log change of on-chain transaction volume. *RV* is daily realized volatility which is calculated as the squared root of the sum of squared 5-minute returns during the day. *Illiq* is the ratio of absolute daily return to trading volume in million US dollars. *VolaLiq* is volatility of liquidity and is calculated following Leirvik (2021). *logTweet* is the natural log of the number of tweets about bitcoin on Twitter. *PC_Network* is the bitcoin network factor constructed following Liu and Tsyvinski (2020) as the first principal component of four bitcoin network measures, including the number of wallet users, the number of active addresses, the number of transaction count, and the number of payment count. *t*-statistics in column (1) are adjusted using Newey-West standard errors with six lags, and in columns (2)–(4) are adjusted using Hodrick (1992) standard errors. *, **, and *** denote significance levels at 10%, 5% and 1% respectively.

	Dependent variable:			
	ret_t (1)	ret_{t+1} (2)	$ret_{[t+1,t+7]}$ (3)	$ret_{[t+1,t+30]}$ (4)
$\Delta Reserve_t$	−0.698*** (−2.931)	−0.497** (−2.370)	−1.429** (−2.516)	−2.430** (−2.085)
ret_t		−0.037 (−1.095)	0.058 (0.683)	0.094 (0.534)
$ret_{[t-21,t-1]}$	0.009* (1.670)	0.011* (1.746)	0.043 (1.011)	0.073 (0.540)
<i>logVolume</i>	0.001 (0.605)	0.002 (1.070)	0.012 (1.226)	0.062 (1.620)
<i>logMktCap</i>	−0.001 (−0.313)	−0.005 (−1.495)	−0.032* (−1.686)	−0.147** (−2.026)
ΔEPU	0.002 (0.875)	−0.002 (−0.977)	0.001 (0.722)	0.004 (1.302)
ΔVIX	−0.063*** (−3.393)	−0.0001 (−0.005)	0.007 (0.149)	0.025 (0.294)
ΔSVI	0.008 (0.765)	0.004 (0.596)	−0.003 (−0.222)	−0.015 (−0.426)
$\Delta WikiSearch$	0.003 (0.818)	0.001 (0.350)	−0.003 (−0.651)	−0.012* (−1.929)
$\Delta TransVol$	0.002 (0.941)	0.001 (0.444)	0.0002 (0.060)	−0.018** (−2.366)
<i>RV</i>	−0.280*** (−2.917)	0.105 (1.382)	0.387 (1.132)	0.183 (0.167)
<i>Illiq</i>	0.971 (0.538)	−0.378 (−0.341)	−5.877 (−1.275)	−23.025 (−1.545)
<i>VolaLiq</i>	0.128 (0.783)	0.006 (0.032)	0.143 (0.112)	2.309 (0.440)
<i>logTweet</i>	0.008*** (3.111)	0.002 (0.514)	0.016 (0.967)	0.112* (1.755)
<i>PC_Network</i>	0.008 (1.494)	−0.005 (−1.080)	0.011* (1.697)	−0.012 (−1.146)
<i>Constant</i>	−0.085* (−1.919)	0.063 (1.384)	0.363 (1.204)	1.095 (0.940)
Observations	1977	1976	1970	1947
R^2	0.051	0.013	0.028	0.059
Adjusted R^2	0.044	0.005	0.020	0.052

addresses.²⁵ Table 15 displays the list of exchanges and Fig. 9 exhibits the aggregate exchange reserves in each data set. The coverage of CryptoQuant and Glassnode are relatively similar with 15 exchanges in common, resulting in nearly identical measures of exchange reserves. Meanwhile, exchanges in CoinMetrics and In-

²⁵ We thank the author of WalletExplorer.com, Aleš Janda, for allowing us to access the exchange address data via the website's API.

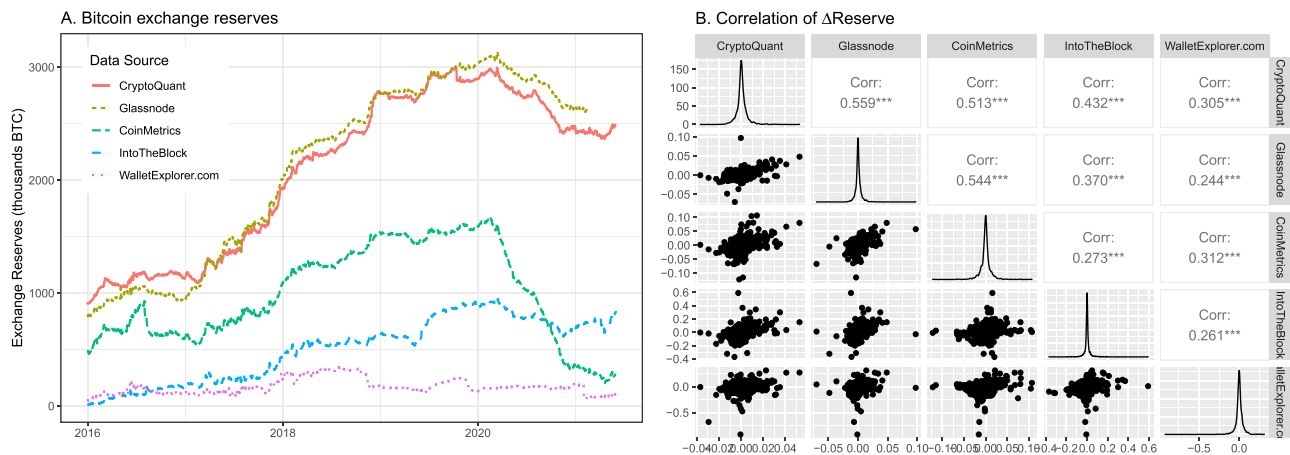


Fig. 9. Bitcoin exchange reserves from alternative data sources. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 14

Out-of-sample forecast on bitcoin returns of competing factors. This table presents Campbell and Thompson's (2008) R^2_{OS} and Clark and West's (2007) z-statistics using one-year, two-year, and three-year training windows. The forecasting model is $ret_{t+1} = \alpha + \beta X_t + \varepsilon_{t+1}$ where X represents the examined factor. Definitions of these factors are presented in Table 13. A positive R^2_{OS} or a statistically significant z-statistics is interpreted as the corresponding factor improving the forecasting accuracy of the benchmark model without the factor. * and ** denote statistical significance at 10% and 5% respectively.

Factor	Criteria	Training window		
		One year	Two years	Three years
$\Delta Reserve$	R^2_{OS} (%)	-0.021	0.211	0.312
	z-statistics	1.042	1.575*	2.059**
$ret_{[t-21,t-1]}$	R^2_{OS} (%)	-0.300	0.100	-0.000
	z-statistics	-0.189	0.885	0.504
$logVolume$	R^2_{OS} (%)	-0.600	-0.800	-0.300
	z-statistics	-0.154	-1.590	0.188
$logMktCap$	R^2_{OS} (%)	-0.600	-0.400	-0.700
	z-statistics	0.197	-0.581	-0.677
ΔEPU	R^2_{OS} (%)	-0.400	-0.100	0.000
	z-statistics	-2.275	-0.544	0.360
ΔVIX	R^2_{OS} (%)	-0.500	-0.400	-0.200
	z-statistics	-1.558	-1.511	-1.164
ΔSVI	R^2_{OS} (%)	-0.400	-0.100	-0.000
	z-statistics	-1.728	-1.368	-0.287
$\Delta WikiSearch$	R^2_{OS} (%)	-0.200	-0.200	-0.200
	z-statistics	0.202	-0.734	-1.244
$\Delta TransVol$	R^2_{OS} (%)	-0.400	-0.200	-0.400
	z-statistics	-1.372	-0.418	-1.250
RV	R^2_{OS} (%)	-0.600	-0.500	-0.300
	z-statistics	-0.570	-0.776	-0.048
$Illiq$	R^2_{OS} (%)	-0.100	-0.100	-0.100
	z-statistics	0.450	-0.230	-0.972
$VolaLiq$	R^2_{OS} (%)	-0.600	-0.600	-0.100
	z-statistics	-0.028	-1.919	-0.581
$logTweet$	R^2_{OS} (%)	-0.400	-0.600	-0.300
	z-statistics	0.323	-0.904	-1.127
$PC_Network$	R^2_{OS} (%)	-0.500	-0.200	-0.000
	z-statistics	-2.597	-1.901	-0.591

toTheBlock are essentially subsets of CryptoQuant and Glassnode. WalletExplorer only has eight exchanges in common with CryptoQuant and provides a significantly smaller estimation of bitcoin reserves. This is not surprising because the website only covers small and inactive exchanges and had stopped updating addresses since 2016, thus the data do not capture the recent development of the bitcoin market.

Using data from five data sources, we also create three combined reserves measures. First, $Combine_{CGW}$ is constructed by merg-

Table 15

Exchanges available in alternative data sources.

The "CryptoQuant" column exhibits the list of exchanges from CryptoQuant which are used in the main analysis. The check mark (✓) indicates that the exchange is also covered by the corresponding alternative data source. Exchanges that are not covered in CryptoQuant but in the alternative data sources are listed in the bottom of the table.

CryptoQuant	Glassnode	CoinMetrics	IntoTheBlock	WalletExplorer
Binance	✓	✓	✓	✓
Bitfinex	✓	✓	✓	✓
BitFlyer				
Bithumb	✓			
BitMex	✓	✓		
Bitstamp	✓	✓	✓	✓
Bittrex	✓	✓		✓
Bybit				
Coinbase Pro	✓		✓	
CoinOne				
FTX	✓			
Gate.io	✓			
Gemini	✓	✓	✓	
Gopax				✓
Huobi	✓	✓	✓	✓
Korbit				✓
Kraken	✓	✓	✓	
Kucoin	✓			
OKEx	✓		✓	
Poloniex	✓	✓	✓	✓
Luno			Luno	79 others ^a
Coincheck				
HitBTC				

^a Refer to www.walletexplorer.com for a comprehensive list of exchanges.

ing exchange reserves from CryptoQuant with that of other non-overlapping exchanges from Glassnode and then from WalletExplorer.com. Second, $Combine_{WCG}$ is constructed in the same manner but the merging starts with WalletExplorer.com, then CryptoQuant and finally Glassnode. A third combined measure is $\Delta Combine_{PC}$, calculated as the first principal component of $\Delta Reserve$ from all five data sources. We then rerun the regression model (1) with $\Delta Reserve$ calculated separately for each data source and for each combined reserves measure. Consistent with the baseline results, Table 16 shows that the regression coefficients of all reserve changes measures are negative and statistically significant except for IntoTheBlock and WalletExplorer.com.²⁶ The changes of re-

²⁶ For IntoTheBlock, the lack of statistical significance could be due to the small number of exchanges covered in the data set. Meanwhile, although WalletExplorer.com covers a large number of exchanges, it had stopped updating exchange

Table 16

Regression results with alternative data sources.

This table reports the regression results of model (1) where $\Delta Reserve$ are calculated using different data sources, including Glassnode, CoinMetrics, IntoTheBlock, and WalletExplorer.com. $\Delta Combine_{CGW}$ is the log changes of the combined reserves measure created by merging reserves of exchanges from CryptoQuant with non-overlapping exchanges from Glassnode and then from WalletExplorer.com. $\Delta Combine_{WCG}$ is constructed in the same manner but starting with WalletExplorer.com, then CryptoQuant and finally Glassnode. $\Delta Combine_{PC}$ is the first principal component of $\Delta Reserve$ from all five data sources including CryptoQuant, Glassnode, CoinMetrics, IntoTheBlock and WalletExplorer.com.

	Dependent variable: ret_{t+1}						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\Delta Glassnode$	-0.279* (-1.835)						
$\Delta CoinMetrics$		-0.185** (-2.225)					
$\Delta IntoTheBlock$			-0.014 (-0.597)				
$\Delta WalletExplorer$				-0.022 (-1.350)			
$\Delta Combine_{CGW}$					-0.672*** (-3.884)		
$\Delta Combine_{WCG}$						-0.198*** (-2.977)	
$\Delta Combine_{PC}$							-0.002*** (-2.673)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1873	1976	1976	1976	1976	1976	1873
R^2	0.009	0.012	0.009	0.010	0.016	0.013	0.012
Adjusted R^2	0.001	0.005	0.001	0.002	0.009	0.005	0.004

serves combined from different data sources better explain future returns as shown by a higher statistical significance of the coefficients and a higher R^2 .

7. Conclusions

This paper examines how bitcoin exchange reserves affect bitcoin prices. Our analysis is based on the hypothesis that changes in exchange reserves imply changes in price pressure, i.e., increased reserves lead to increased selling pressure and decreased reserves lead to decreased selling pressure. Consistent with this notion, we find that bitcoin transfers onto exchanges (from investors' private wallets) result in increased bitcoin exchange reserves and negatively affect bitcoin prices on the same day and one-day ahead. Transfers off exchanges (to investors' private wallets) result in decreased bitcoin exchange reserves and positively affect long-term bitcoin prices.

We also investigate triggers of exchange reserve transfers and find that an extreme fall of bitcoin prices leads investors to transfer bitcoin from private wallets to exchange accounts implying further selling. An increase in bitcoin volatility has the same trigger effect during periods of normal volatility, but not during periods of extremely low or extremely high volatility. The trigger effect and the differences across volatility regimes are well aligned with the "loaded for bear" analogy - increased risk triggers the loading of the gun but only if the level of risk is not too low or already too high.

We also demonstrate that although exchanges are involved in more than 80% of on-chain transactions, a significant fraction of bitcoin investors do not leave their bitcoin on exchanges, indicating that they prefer to store their wealth off exchanges in private wallets, and suggesting that bitcoin investors have more trust in the blockchain and its wallets than in exchanges. This is consistent with the security issues of cryptocurrency exchanges (Moore

and Christin, 2013; Bhaskar and Chuen, 2015), and with the fact that centralized exchanges can be viewed as "central authority" or "middleman" which is what the blockchain technology aims to eliminate (Nakamoto, 2008).

This study highlights a fundamental difference between crypto trading and traditional financial markets. The crypto space allows to store assets privately in private wallets whereas investors in traditional assets such as stocks or fiat currencies do not have such option but have to store their assets in their brokers' nominee accounts. Since the movement of crypto assets across wallets are observable thanks to the transparency of the blockchain technology, this difference offers opportunities for further research on investors' trust and trading behavior. Since the crypto industry uses a variety of indicators and measures and not only exchange reserve changes, future research could also analyze other indicators and measures.

CRedit authorship contribution statement

Lai T. Hoang: Conceptualization, Methodology, Software, Data curation, Visualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Dirk G. Baur:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

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Appendix A

Consider the baseline regression model (1):

$$ret_{[t+k,t+h]} = \alpha + \beta \Delta Reserve_t + \sum \gamma_i Control_{i,t} + \varepsilon_{[t+k,t+h]} \quad (10)$$

addresses since 2016, thus a majority of those exchanges are small or inactive. Further examination shows that for the last three years of the sample period, WalletExplorer.com's exchange reserves are mostly attributed to the eight exchanges that overlap with the CryptoQuant data.

Applying generalized method of moments (GMM), Hodrick (1992) shows that the coefficients $\theta = (\alpha \ \beta \ \gamma_i)'$ have an asymptotic distribution $\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Omega)$ where $\Omega = Z_0^{-1} S_0 Z_0^{-1}$, $Z_0 = E(x_t x_t')$, and $x_t' = (1 \ \Delta Reserve_t \ Control_{i,t})$. $\hat{S}_0$ is estimated as:

$$\hat{S}_0 = \frac{1}{T} \sum_{t=k}^T w k_t w k_t' \quad (11)$$

where $w k_t = \epsilon_{t+1} \left(\sum_{i=0}^{k-1} x_{t-i} \right)$, and T is the number of observations.

Let $\hat{\beta}_k$ denote the estimated coefficient of $\Delta Reserve$ in model (1) where $ret_{[t+1, t+h]}$ is the dependent variable, then Boudoukh et al. (2019) standard errors are calculated as:

$$se(\hat{\beta}_h) = se(\hat{\beta}_h^{nol}) \times \sqrt{\frac{1}{h} + 2 \sum_{j=1}^{h-1} \frac{h-j}{h^2} \rho_j} \quad (12)$$

where $\hat{\beta}_h^{nol}$ is the estimated coefficient of $\Delta Reserve$ using a non-overlapping sample, i.e., a new observation is added after every h days, and ρ_j is the j th-order autocorrelation of $\Delta Reserve_t$.

Stambaugh's (1999) adjusted standard error for the one-day-ahead predictive regression is calculated as follows. We first estimate the following models using Ordinary Least Squares:²⁷

$$ret_{t+1} = \alpha + \beta \Delta Reserve_t + u_t \quad (13)$$

$$\Delta Reserve_{t+1} = \theta + \rho \Delta Reserve_t + v_t \quad (14)$$

and obtain the estimated coefficients ($\hat{\beta}$ and $\hat{\rho}$) and residuals ($\hat{u}_t$ and $\hat{v}_t$). Then the corrected coefficient of the predictive model (13) is calculated as follows:

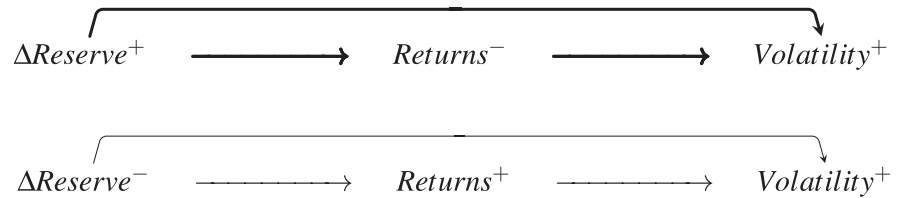
$$\beta^* = \hat{\beta} - \frac{\sigma_{uv}}{\sigma_v^2} \times \frac{1 + 3\hat{\rho}}{T} \quad (15)$$

where σ_{uv} is the covariance between $\hat{u}_t$ and $\hat{v}_t$, σ_v^2 is the variance of $\hat{v}_t$, and T is the number of observations.

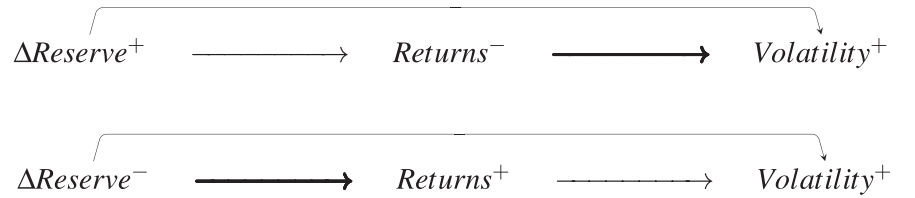
Appendix B

This Appendix provides an explanation for short-term asymmetric and long-term insignificant effects of bitcoin exchange reserve changes ($\Delta Reserve$) on bitcoin volatility. We demonstrate that the results can be explained with the results on the effect of $\Delta Reserve$ on bitcoin returns and the “classical” asymmetric volatility or leverage effect where negative returns increase volatility by more than positive returns (Glosten et al., 1993). Specifically, on the left hand side, thick arrows represent the significant effects of positive reserve changes on short-term returns and of negative reserve changes on long-term returns as shown in Table 4. The right-hand-side arrows illustrate the leverage effect where the stronger effect of negative returns on volatility is represented by thick arrows. In all cases, thin arrows indicate insignificant effects. For short horizons, significant effects on both sides lead to a total significant effect, which is represented by the thick overhead arrow in the upper part. Insignificant effects on either side lead to a total insignificant effect as illustrated by thin overhead arrows in the remaining cases.

Short term asymmetric effect



Long term insignificant effect



²⁷ In contrast to the baseline model (1), control variables are not included in model (13) as the bias can no longer be signed in a multivariate regression (Ang and Bekaert, 2007; Stambaugh, 1999).

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